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Kolmogorov-Smirnov type statistics for structural breaks and parameter constancy - a new adjusted-range based self-normalization approach

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Abstract

The existing self-normalization approach in time series utilizes the variance of a partial sum process as the self-normalizer, which is more sensitive to irregularities such as persistent autocorrelation, heteroskedasticity, unit root and outliers. In this paper we propose a self-normalization approach based on the adjusted-range of a partial sum, which is robust to those irregularities and helps to rectify the "better size but less power" phenomenon (Wang and Shao, 2020) for the self-normalized test statistics using the existing self-normalization approach. We also introduce adjusted-range based Kolmogorov-Smirnov (KS) type statistics to test for structural breaks in the mean of a univariate time series, in approximately linear statistics (general context) and parameter constancy in a regression setting. Our approach can rectify the well known power decrease issue associated with self-normalized KS tests without the use of backward and forward summation as in the G statistic of Shao and Zhang (2010), and circumvents the specification of a contrast process as in Zhang and Lavitas (2018). Furthermore, our proposed test statistics are portmanteau and can cater for general alternatives. Finally, Monte Carlo simulations and empirical studies demonstrate the adequacy of our proposed method.

Keywords: Change-point testing; CUSUM process; Parameter constancy; Portmanteau test; Studentization.

1 Introduction

This paper proposes the adjusted-range based self-normalization approach, and applies it to testing for structural breaks and parameter constancy in a time series context. We thus contribute to two pivotal domains of statistical/econometric research, viz. self-normalization and structural break tests.

The self-normalization approach is employed to avoid the estimation for the long-run variance (LRV). In general, valid statistical inference often needs accurate estimation of the LRV, which, in turn, requires modelling the rate of decay of the autocovariance function. Substantial effort has been invested in providing suitable estimates of the LRV. Perhaps the most famous method is the so-called heteroskedasticity and autocorrelation consistent (HAC) LRV estimator espoused by White (1980), Newey and West (1987, 1994), Andrews (1991) and Andrews and Monahan (1992). However, authors such as Andrews (1991) and Den Haan and Levin (1997) found that the small sample performance of these HAC LRV estimators was poor. Müller (2007) suggested that these asymptotically consistent HAC LRV estimators, despite being theoretically appealing and empirically simple to apply, often lead to tests with bad sizes in small samples with realistic amounts of dependence.

Another approach is to use the so called fixed-b asymptotic method, proposed by Kiefer et al. (2000), Kiefer and Vogelsang (2002) and Kiefer and Vogelsang (2005). Instead of assuming that the bandwidth b goes to zero as the sample size approaches infinity, under fixed-b asymptotics, $b \in (0, 1]$ is set to be a fixed number. In standard asymptotics, such a b is relatively very small for large sample sizes; perhaps for this reason, Kiefer and Vogelsang (2005) referred to the standard asymptotics as small-b asymptotics. Kiefer and Vogelsang (2005) outlined two advantages of the fixed-b approach. First, fixed-b asymptotics enables a more accurate first order approximation for HAC robust tests. Second, fixed-b asymptotic theory permits a local asymptotic power analysis for HAC robust tests. The fixed-b asymptotics can enhance the small sample performance of various tests (Kiefer et al., 2000; Kiefer and Vogelsang, 2002, 2005). However, the fixed-b approach has a trade-off between size distortion and power loss; for larger b the size distortion rectifies itself but the power loss increases (Kiefer and Vogelsang, 2005; Shao, 2015). Clearly, the asymptotic distribution of the fixed-b based HAC tests depends on the parameter b; but there is, however, no general guidance on choosing b, and it must be specified in some ad hoc fashion. Moreover, Shao (2015) reviewed the Monte Carlo simulation results from Kiefer and Vogelsang (2005), Shao (2010) and Shao and Zhang (2010) and found that when the series is a unit-root or near unit-root process, the size of the fixed-b based HAC tests deteriorates.

Fortunately, it is possible to circumvent LRV estimation and still conduct sound statistical tests through the self-normalization approach. The concept of self-normalization is not new. It originates with Student (1908)[1], whose widely applied and celebrated t statistic and t-distribution first utilized this concept. Despite the fact that the sample variance based on a small number of

[1]Student was a pseudonym of William Sealy Gosset working under commercial secrecy at the Guinness Brewery in Dublin.

observations is an inadequate estimator of the population variance, it is stochastically proportional to the population variance and can be used as a normalizer. The concept of self-normalization, has become an important principle in conducting statistical inference (Shao, 2015).

The self-normalized approach to inference for time series was first introduced in Shao (2010) as an extension of an idea devised and developed by Kiefer et al. (2000) and Lobato (2001). They formulated an inconsistent LRV estimator based on the sample variance of a partial sum. Despite the inconsistency, such a LRV estimator is stochastically proportional to the true LRV and is nuisance parameter free. Since its introduction, self-normalization has been deployed in various aspects of statistical inference (Shao, 2015). To summarize, the self-normalization proposed by Lobato (2001) and Shao (2010) has been deployed in statistical inference for issues such as confidence interval construction (Shao, 2010), tests for autocorrelation (Lobato, 2001; Shao, 2010; Boubacar-Maïnassara and Saussereau, 2018), tests for structural breaks (Shao and Zhang, 2010; Zhang et al., 2011; Zhang and Lavitas, 2018) and for the handling of various types of data, such as functional time series (Zhang et al., 2011; Dette et al., 2020), spatial data (Zhang et al., 2014), censored dependent data (Huang et al., 2015) and alternating regime index datasets (Kim and Shin, 2020). Self-normalization has also been applied across many academic fields of study, for example economics (Lobato, 2001; Shao, 2010), finance (Choi and Shin, 2020a,b), ecology (Zhang et al., 2014), climate studies (Dette et al., 2020) and epidemiology (Jiang et al., 2020).

The self-normalization proposed by Lobato (2001) and Shao (2010) is based on the variance of the partial sum of a process, which is sensitive to irregularities such as persistent autocorrelation, heteroskedasticity, unit root and outliers. In place of using the sample variance, we propose the use of the adjusted range of a partial sum. Similar to the work of Lobato (2001) and Shao (2010), the sample range of a partial sum is asymptotically proportional to the square root of the LRV up to a stochastic factor and its distribution is nuisance parameter free, hence it can be used as a valid self-normalizer. Moreover, range has some appealing robustness properties; specifically, the ability to deal with persistent autocorrelation and to accommodate irregularities such as outliers, high levels of skewness/kurtosis and unit-root behavior in volatilities. For example, Mandelbrot and Wallis (1969) showed by Monte Carlo simulation studies that a range statistic can effectively detect long-range dependence in highly non-Gaussian time series with large skewness and/or kurtosis. Mandelbrot (1972, 1975) showed the almost-sure convergence of the range statistic for stochastic processes with infinite variances, a distinct advantage over the methods based on autocorrelations and variance ratios which may not be well-defined for infinite variance processes. Moreover, the range as a statistical quantity has also been widely applied in financial volatility estimation (Chou et al., 2010).

A cornerstone of statistical/econometric analysis is the structural stability of the data/model under consideration and a failure to incorporate structural breaks will inevitably lead to unreliable inferences and forecasts. The importance of testing for structural breaks is also reflected in the huge amount of related literature focusing on various contexts. See Hansen (2001) and Aue and Horváth (2013) for literature reviews on testing for structural breaks. Depending on the quantities

of interest, structural break tests can be conducted on the mean, variance or covariance structure and for general model stability. See, for example, Stock and Watson (1996), Bai (1997), Altissimo and Corradi (2003) and Harchaoui and Lévy-Leduc (2010) for references on testing for structural breaks in the mean. Testing for a structural break in the variance is particularly important in finance and economics, where the stability of process volatility is a crucial issue; for references see, for instance, Inclan and Tiao (1994), Chen and Gupta (1997) and Smith (2008). There are also efforts combining tests for structural breaks in the mean and in the variance, for instance, Wang and Zivot (2000), Pitarakis (2004) and Jin et al. (2018). Moreover, Aue et al. (2009) tested for testing structural breaks in the covariance structure.

There are also substantial efforts devoted to testing breaks in regression models. Noticeably, the constancy of parameter tests are shown to be equivalent to tests of the mean of certain residuals, such as the estimated one-step ahead prediction errors. In particular, since Brown et al. (1975) introduced the cumulative sum (CUSUM) test based on recursive residuals, a large number of tests for parameter constancy based on CUSUM processes have been formulated. For instance, Krämer et al. (1988) considered the CUSUM test for structural breaks when lagged dependent variables are incorporated in the linear model, Krämer et al. (1988) showed that the CUSUM tests retain their asymptotic significance levels in dynamic models. Ploberger and Krämer (1992) considered the use of a CUSUM test when testing for parameter constancy in a linear model. See Andrews (1993), Bai and Perron (1998), Perron and Qu (2007) and Qu and Perron (2007) for more studies on testing breaks in regression models.

Our contribution is three-fold. First, we introduce the adjusted-range based self-normalization in time series, in particular, we extend the univariate adjusted-range based LRV estimator to the adjusted-range based covariance estimator through developing a three-step procedure. First, this procedure transforms the series through a "square-root-free Cholesky decomposition", which gathers all the non-trivial covariances to the diagonal component of the long-run covariance estimator; then we obtain a diagonal adjusted-range based covariance estimator, which is subsequently converted to that of the original series through reverting the "square-root-free Cholesky decomposition", see Section 3 for detail. The procedure bears similarity to the structural vector autoregression (SVAR) literature and VAR prewhitening of Andrews and Monahan (1992), and is robust to irregularities such as persistent autocorrelation, heteroskedasticity, unit root, outliers, and so on. Second, we find, through simulation studies, that our adjusted-range self-normalization approach helps to moderate the "better size but less power" phenomenon (Wang and Shao, 2020) exhibited by the existing self-normalized test statistics of Lobato (2001) and Shao (2010). In particular, the adjusted-range based structural break tests can achieve good sizes while maintaining reasonable powers, especially for multivariate series. Furthermore, they can ameliorate the poor finite sample performance of the HAC based tests (Müller, 2007), help to rectify the nonmonotonic power issue without the need for forward and backward summation, as employed by Shao and Zhang (2010) when introducing the G test, and enable us to circumvent the specification of a contrast process as in Zhang and Lavitas (2018). Third, our proposed adjured-range based test statistics are portmanteau tests and hence

potentially powerful against a range of alternative hypotheses, such as smooth structural change. For abrupt changes, the number of break points does not need to be specified a priori in any ad hoc fashion. This is in contrast to the G tests proposed by Shao and Zhang (2010), which must be formulated with a pre-specified number of change-points.

The remainder of the article is organized as follows. Section 2 considers testing for change in the mean of a process and introduces the adjusted-range based self-normalization in the univariate case. Section 3 introduces the adjusted-range based self-normalization in a multivariate context, and covers testing for change in a general framework, viz. the approximately linear statistics, which includes the marginal mean, the marginal variance, the autocorrelation function, quantiles and the spectrum as special cases. Section 4 introduces the adjusted-range based tests for parameter constancy. Sections 5 and 6 conduct simulation studies and empirical analysis, respectively. Finally, conclusions are drawn in Section 7.

2 Testing for structural breaks in the mean and the adjusted-range based self-normalization

This section considers testing for structural breaks in the mean of a univariate time series, and introduces the adjusted-range based self-normalization in a univariate context. Our focus on the mean is motivated by the arguments of Carlstein et al. (1994), Brodsky and Darkhovsky (2005) and Harchaoui and Lévy-Leduc (2010), who suggested that detecting structural changes in a time-evolving statistical quantity can generally be reduced to detecting changes in the mean of an alternative sequence derived from the initial one.

Consider a univariate series $\{X_t\}$, we wish to test the hypothesis

$$\mathbb{H}_0 : E(X_1) = \cdots = E(X_n) = \mu, \tag{1}$$

versus

$$\mathbb{H}_1 : \text{otherwise}. \tag{2}$$

Define $\overline{X}_n = n^{-1}\sum_{t=1}^{n} X_t$ and the CUSUM process as

$$\mathrm{T}_n(\lfloor sn \rfloor) = n^{-1/2}\sum_{t=1}^{\lfloor sn \rfloor}\left(X_t - \overline{X}_n\right), \tag{3}$$

where $s \in [0,1]$ and $\lfloor \cdot \rfloor$ denotes the integer part. Under appropriate moment and weak dependence conditions, as in for instance, Assumption 2.1 in Phillips (1987) (also see Assumptions 1-4 below)

$$n^{-1/2}\sum_{t=1}^{\lfloor sn \rfloor}\left[X_t - E(X_t)\right] \Rightarrow \sigma B(s), \tag{4}$$

where $B(s)$ is the one-dimensional Brownian motion (Wiener process), σ^2 is the true variance of

$\{X_t\}$ and "$\Rightarrow$" denotes weak convergence. Under the null $\mathbb{H}_0$, we have $\mathrm{T}_n(\lfloor sn \rfloor) \Rightarrow \sigma \mathbb{B}(s)$, where $\mathbb{B}$ is a Brownian Bridge and $\mathbb{B}(s) = B(s) - sB(1)$.

The CUSUM statistic, also known as the KS statistic satisfies,

$$\mathrm{KS}_n = \sup_{s \in [0,1]} \left| \mathrm{T}_n(\lfloor sn \rfloor) / \sigma \right| \xrightarrow{d} \sup_{s \in [0,1]} \left| \mathbb{B}(s) \right|, \tag{5}$$

where "$\xrightarrow{d}$" denotes convergence in distribution. Clearly, if we could obtain a consistent estimate, $\widehat{\sigma}_n$, for σ, then by the continuous mapping theorem (CMT), we have

$$\mathrm{KS}_n^0 = \sup_{s \in [0,1]} \left| \mathrm{T}_n(\lfloor sn \rfloor) / \widehat{\sigma}_n \right| \xrightarrow{d} \sup_{s \in [0,1]} \left| \mathbb{B}(s) \right|. \tag{6}$$

Here we use KS_n^0 to denote the KS statistic evaluated under the Bartlett kernel and the small-b asymptotics. Usually, consistent estimates for σ^2 are obtained through kernel-based LRV estimation using standard asymptotics, albeit with poor finite sample performance (Andrews, 1991; Den Haan and Levin, 1997).

Alternatively, the fixed-b approach could be deployed, under suitable conditions and using the Bartlett kernel, the LRV estimator based on the fixed-b asymptotics satisfies $\widehat{V}(b) \Rightarrow Q(b)\sigma^2$, where $Q(b) = \frac{2}{b}\left[\int_0^1 [\mathbb{B}(s)]^2 ds - \int_0^{1-b} \mathbb{B}(s+b)\mathbb{B}(s) ds\right]$. Clearly, by CMT, we have

$$\mathrm{KS}_n^{(b)} = \max_{1 \le k \le n} \left| \widehat{V}(b)_n^{-1/2} \mathrm{T}_n(k) \right| \xrightarrow{d} \frac{\sup_{s \in [0,1]} \left| \mathbb{B}(s) \right|}{\sqrt{Q(b)}}. \tag{7}$$

To circumvent LRV estimation altogether, Lobato (2001) and Shao (2010) proposed the self-normalization approach. Note that the self-normalization approach in Shao (2010) can be viewed as a special case of the fixed-b asymptotics in Kiefer and Vogelsang (2005), when $b = 1$ and the kernel is the Bartlett kernel. Shao and Zhang (2010) considered its use in testing for the presence of structural breaks. Define $\overline{X}_t = t^{-1}\sum_{i=1}^t X_i$ and $W_n^2 = n^{-2}\sum_{t=1}^n t^2 \left(\overline{X}_t - \overline{X}_n\right)^2$. By the functional central limit theorem (FCLT), we have $W_n^2 \Rightarrow \sigma^2 \int_0^1 [\mathbb{B}(s)]^2 ds$. Using the CMT, we can easily obtain

$$\mathrm{KS}_n^W = \max_{1 \le k \le n} \left| W_n^{-1} \mathrm{T}_n(k) \right| \xrightarrow{d} \frac{\sup_{s \in [0,1]} \left| \mathbb{B}(s) \right|}{\int_0^1 [\mathbb{B}(s)]^2 ds}. \tag{8}$$

KS_n^W (8) is referred to as the "naive extension to the self-normalization idea" by Shao and Zhang (2010). More specifically, Shao and Zhang (2010) introduced a shift in a moderately correlated time series, and found that the percentage of rejections gradually decreases as the magnitude of change increases, which is also related to the well-known phenomena of nonmonotonic power when testing for structural breaks. Shao and Zhang (2010) attributed this result to the increase in the denominator of KS_n^W (i.e. W_n) with respect to the break size. To address this issue, Shao and

Zhang (2010) introduced the G test statistic.

Maintain the null hypothesis (1), and consider a more specific alternative

$$\mathbb{H}_1 : E(X_1) = \cdots = E(X_{k^*}) \neq E(X_{k^*+1}) = \cdots = E(X_n), \tag{9}$$

where k^* is the unknown location of the structural break, and $1 \leq k^* < n$.

Define the CUSUM process $S_{t_1,t_2} = \sum_{i=t_1}^{t_2} X_i$ if $t_1 \leq t_2$, and $S_{t_1,t_2} = 0$ otherwise; and the self-normalization process $V_n(\cdot)$

$$V_n(k) = \frac{1}{n}\left\{\sum_{t=1}^{k}\left[S_{1,t} - \frac{t}{k}S_{1,k}\right]^2 + \sum_{t=k+1}^{n}\left[S_{t,n} - \frac{n-t+1}{n-k}S_{k+1,n}\right]^2\right\}, \tag{10}$$

for $k = 1, \ldots, n$. Clearly, $S_{1,t}$ denotes the forward summation, whereas $S_{t,n}$ the backward summation. The G test statistic is

$$\mathrm{G}_n = \max_{1\leq k\leq n-1} \mathrm{T}_n(k)^{\mathrm{T}} V_n^{-1}(k)\, \mathrm{T}_n(k). \tag{11}$$

Provided that (4) remains valid, then following Shao and Zhang (2010) we have

$$\mathrm{G}_n \xrightarrow{d} \sup_{s\in[0,1]} \mathbb{B}(s)^{\mathrm{T}} V^{-1}(s)\, \mathbb{B}(s), \tag{12}$$

where

$$V(s) = \int_0^s \left[B(k) - \frac{k}{s}B(s)\right]^2 dk + \int_s^1 \left\{B(1) - B(k) - \frac{1-k}{1-s}\left[B(1) - B(s)\right]\right\}^2 dk. \tag{13}$$

Intuitively, the self-normalizer $V_n(k)$ accommodates both the forward partial sum before k^* and backward partial sum after k^*, and so is invariant with respect to the structural shift, viz. $\Delta_n := E(X_{k^*+1}) - E(X_{k^*})$. The asymptotic critical values for G_n can also be obtained via simulations, and are summarized as Table 1 in Shao and Zhang (2010). Clearly, the G_n test statistic is better suited to testing if there is one single change point (9), because the supremum in (11) can only be attained at k^*, where the structural shift is located. Shao and Zhang (2010) also considered a multiple change-point alternative, which requires pre-specification of the number of change points and further complicates the specification of the test. Shao and Zhang (2010) suggested that the number of change points can be estimated through treating change point estimation and testing as model selection, see, for example,Yao (1988), Davis et al. (2006), Zhang and Siegmund (2007). Alternatively, one can adopt a sequential testing procedure, see, for instance, Bai and Perron (1998), Qu and Perron (2007) and the references therein.

2.1 Adjusted-range based self-normalization

For a univariate series $\{X_t\}$, the adjusted range is

$$\widehat{R}_n = \max_{1\leq k\leq n} n^{-1/2}\sum_{t=1}^{k}\left(X_t - \overline{X}_n\right) - \min_{1\leq k\leq n} n^{-1/2}\sum_{t=1}^{k}\left(X_t - \overline{X}_n\right). \tag{14}$$

Clearly, $\widehat{R}_n \geq 0$. Such non-negativity means that extension of adjusted-range based self-normalization to the multivariate context is not straightforward, see Section 3 for the details.

To derive the asymptotic distribution of $\widehat{R}_n$ and the adjusted-range based test statistics, we assume that $X_t = \mu + \varepsilon_t$, where μ is a fixed, yet unknown parameter, and ε_t is a zero-mean time series process. Following Phillips (1987) and Lo (1991), we impose the following assumptions on ε_t.

Assumption 1. $E(\varepsilon_t) = 0$ *for all* t*;*

Assumption 2. $\sup_t E\left(|\varepsilon_t|^{2\beta}\right) < \infty$ *for some* $\beta > 2$*;*

Assumption 3. $0 < \sigma^2 = \lim_{n\to\infty} E\left[n^{-1}\left(\sum_{t=1}^{n}\varepsilon_t\right)^2\right] < \infty$*;*

Assumption 4. $\{\varepsilon_t\}$ *is a strong-mixing process with mixing coefficients* γ_k *that satisfy* $\sum_{k=1}^{\infty}\gamma_k^{1-2/\beta} < \infty$*.*

Assumptions 1-4 are also identical to Assumption 2.1 in Phillips (1987). Moreover, these assumptions are imposed by Lo (1991) in the study of long-term memory of stock market prices. See Phillips (1987) for discussions on these assumptions. [2]

Under Assumptions 1-4, we have

$$\widehat{R}_n \Rightarrow \sigma\left[\sup_{s\in[0,1]}\mathbb{B}(s) - \inf_{s\in[0,1]}\mathbb{B}(s)\right]. \tag{15}$$

Combing (15) with (5), and using the CMT, we obtain

$$\mathrm{KS}_n^R = \max_{1\leq k\leq n}\left|\widehat{R}_n^{-1}\mathrm{T}_n(k)\right| \xrightarrow{d} \frac{\sup_{s\in[0,1]}|\mathbb{B}(s)|}{\sup_{s\in[0,1]}\mathbb{B}(s) - \inf_{s\in[0,1]}\mathbb{B}(s)} \tag{16}$$

under the null hypothesis. Note, KS_n^R is suitable for testing the more general alternative (2).

[2] Note that the earlier theoretical results on the partial range pre-dates Phillips (1987) and Lo (1991). Specifically, the asymptotic distribution of the partial sum of independently and identically distributed (i.i.d.) sequences was first derived by Feller (1951) and Donsker (1951). While, Ibragimov (1962) and Billingsley (1968) derived the asymptotic distributions of the partial sum and the range for stationary time series under ψ-mixing conditions. Phillips (1987) and Lo (1991) derived the asymptotic distributions of the partial sum and the range of the partial sum for a time series process with α-mixing condition, which allows certain degrees of weak temporal dependence and heterogeneity. For that reason, we imposed the conditions from Phillips (1987) and Lo (1991), due to the prevalence of temporal dependence and heterogeneity in time series analysis.

The asymptotic distributions of KS_n^W, $\mathrm{KS}_n^{(b)}$ and KS_n^R are nonstandard. However, they are nuisance parameter-free and can be obtained through Monte Carlo simulations. Following Kiefer et al. (2000) and Lobato (2001), we use $10,000$ Monte Carlo simulations and the Brownian motion is approximated by the normalized sum of a sequence of independently and identically distributed (i.i.d.) increments using $200,000$ $N(0,1)$ variables. Note that the results for KS_n are approximated by KS_n^0 here. The asymptotic critical values for various KS test statistics are summarized in Table 1. The histograms and empirical distributions for KS_n^R, KS_n^0, KS_n^W and $\mathrm{KS}_n^{(b)}$ are visualized in Figure 1. Clearly, the smaller the b, the less dispersed the distribution for $\mathrm{KS}_n^{(b)}$ becomes. The distribution for $\mathrm{KS}_n^{(b)}$ gets closer to that of KS_n^W as b becomes larger. When b is small/moderate ($b = 0.1$ or 0.5), the distributions for the $\mathrm{KS}_n^{(b)}$ are similar to that of KS_n.

[Insert Table 1 here.]

[Insert Figure 1 here.]

We investigate the same data generating process (DGP) as in Shao and Zhang (2010). Upon uncovering a nonmonotonic power problem for KS_n^W, Shao and Zhang (2010) used this case as the motivation for introducing the G statistic. The DGP is as follows. Let $u_t = 0.5u_{t-1} + \varepsilon_t$, $\varepsilon_t \sim$ i.i.d. $N(0,1)$, $t = 1, ..., n$ and let

$$y_t = \begin{cases} u_t, \ 1 \leq t \leq n/2 \\ \eta + u_t, \ n/2 + 1 \leq t \leq n \end{cases}, \tag{17}$$

where $\eta \geq 0$. We specify 101 possible values for η through dividing the interval $[0, 6]$ into an equally spaced grid, such that $\eta \in \{0.00, 0.06, 0.12 \ldots, 6.00\}$. We then evaluate the performances of G_n and KS_n^R. The sample size is set to be $n = 200$ as in Shao and Zhang (2010). From Figure 2, it is clear that both G_n and KS_n^R offer increasing power as the magnitude of change increases. Thus, KS_n^R can rectify the nonmonotonic power issue of KS_n^W without the need for forward and backward summation as in the G test. It can circumvent the construction of the contrast process as in Zhang and Lavitas (2018). Because KS_n^R is a portmanteau test, it is potentially powerful against a range of alternative hypotheses. All these advantages suggest the use of the adjusted-range based self-normalization approach in testing for structural breaks, due to its simplicity and potential power against more general alternatives.

[Insert Figure 2 here.]

3 Testing for structural breaks in a general framework and multivariate extension of the adjusted-range based self-normalization

Similar to Shao (2010), we extend the KS test statistic for the mean to a more general setting. We shall show that, under suitable conditions, testing for structural breaks in a general quantity of

interest can be achieved through constructing an adjusted-range based multivariate KS statistic, named as the extended KS test statistic (EKS).

Here we adopt the same framework as Shao (2010) and Shao and Zhang (2010). Let $\mathbf{F}^m$ denote the m-dimensional marginal distribution of X_t, where m is a fixed but arbitrary dimension parameter. Let $\mathbf{Y}_t = (X_t, \ldots, X_{t+m-1})^{\mathrm{T}}$, $t = 1, \ldots, N = n - m + 1$, and denote $\mathbf{F}_t^m$ as the distribution of $\mathbf{Y}_t$. Let

$$\theta_t = \mathbf{T}\left(\mathbf{F}_t^m\right) \in \mathbb{R}^q,\ t = 1, \ldots, N, \tag{18}$$

be the quantity of interest, where $\mathbf{T}$ is a functional that takes value in $\mathbb{R}^q$. We are interested in testing

$$\mathbb{H}_0 : \theta_1 = \cdots = \theta_N, \tag{19}$$

versus

$$\mathbb{H}_1\text{: } \mathbb{H}_0 \text{ is untrue;}$$

$\mathbb{H}_1$ indicates that there exists at least one k^*, such that $\theta_{k^*} \neq \theta_{k^*+1}$, where $1 \leq k^* < N$. Examples of θ include, but are not limited to, the marginal mean of X_t, the marginal variance of X_t, the autocorrelation function (ACF) of X_t and the quantiles of the distribution of $\mathbf{F}^1$ (Shao and Zhang, 2010).

Usually, $\mathbf{F}_t^m$ is unknown, though we can estimate it using the empirical distribution function. Let ρ_{t_1,t_2} be the empirical distribution based on $\{\mathbf{Y}_j\}_{j=t_1}^{t_2}$:

$$\rho_{t_1,t_2} = (t_2 - t_1 + 1)^{-1} \sum_{t=t_1}^{t_2} \delta_{\mathbf{Y}_t}, \tag{20}$$

where $\delta_{\mathbf{y}}$ is the probability measure which puts mass 1 at point $\mathbf{y}$; see Definition 1 in Hampel et al. (p.84, 1986). Clearly, $\rho_{1,N}$ is the empirical distribution based on $\{\mathbf{Y}_j\}_{j=1}^{N}$.

Following Kunsch (1989), we consider an approximately linear statistic:

$$\widehat{\theta}_{1,N} = \mathbf{T}\left(\rho_{1,N}\right) = \mathbf{T}\left(\mathbf{F}^m\right) + N^{-1} \sum_{t=1}^{N} \mathbf{IF}\left(\mathbf{Y}_t; \mathbf{F}^m\right) + \mathbf{R}_{1,N}, \tag{21}$$

where $\mathbf{IF}\left(\mathbf{Y}_t; \mathbf{F}^m\right)$ is the influence function of $\mathbf{T}$ at $\mathbf{F}^m$, such that

$$\mathbf{IF}\left(\mathbf{y}; \mathbf{F}^m\right) = \lim_{\epsilon \downarrow 0} \frac{\mathbf{T}\left\{(1-\epsilon)\,\mathbf{F}^m + \epsilon \delta_{\mathbf{y}}\right\} - \mathbf{T}\left(\mathbf{F}^m\right)}{\epsilon}, \tag{22}$$

and $\mathbf{R}_{1,N}$ is the remainder term of the expansion. Note that Kunsch (1989) imposes a set of assumptions to ensure the validity of the expansion in (21); specifically, Kunsch (1989) assumed the almost sure convergence of $\mathbf{T}\left(\rho_{1,N}\right)$ to $\mathbf{T}\left(\mathbf{F}^m\right)$, and the existence of the influence function $\mathbf{IF}\left(\mathbf{y}; \mathbf{F}^m\right)$ for all $\mathbf{y} \in \mathbb{R}^m$, denoted as (A1) and (A2) respectively in Kunsch (1989). By focusing on the class of approximately linear statistics, we implicitly assume that Assumptions (A1) and (A2) in Kunsch (1989) hold.

Note that the expansion Kunsch (1989) considered in (21) is based on the entire sample, whereas our method relies on estimation based on a subsample, i.e. $\left(\mathbf{Y}_1, \mathbf{Y}_2, \ldots, \mathbf{Y}_{\lfloor sN \rfloor}\right)$. Thus, we work with the expansion in the subsample (23), and develop our asymptotic results based on Lemma 6 and Assumption 6 instead. Similar to Shao and Zhang (2010), for $1 \leq t_1 \leq t_2 \leq N$, we have

$$\widehat{\theta}_{t_1,t_2} = \mathbf{T}\left(\rho_{t_1,t_2}\right) = \mathbf{T}\left(\mathbf{F}^m\right) + \left(t_2 - t_1 + 1\right)^{-1} \sum_{t=t_1}^{t_2} \mathbf{IF}\left(\mathbf{Y}_t; \mathbf{F}^m\right) + \mathbf{R}_{t_1,t_2}. \tag{23}$$

We define a process based on $\widehat{\theta}_{t_1,t_2}$, i.e.

$$\mathbf{T}_n\left(k\right) = kN^{-1/2}\left(\widehat{\theta}_{1,k} - \widehat{\theta}_{1,N}\right), \tag{24}$$

where $\widehat{\theta}_{1,k}$ is based on $\{\mathbf{Y}_j\}_{j=1}^{k}$, and $\widehat{\theta}_{1,N}$ on $\{\mathbf{Y}_j\}_{j=1}^{N}$, which is the entire sample. The expression for $\mathbf{T}_n\left(k\right)$ is similar to (3), in that $\overline{X}_N$, being the sample mean, uses the whole sample, while $k\widehat{\theta}_{1,k}$ is the analogue of the term $\sum_{t=1}^{\lfloor sn \rfloor} X_t$ in (3), which uses only part of the sample.

Following Shao and Zhang (2010), let $\mathcal{D}\left[0,1\right]$ be the space of functions on $[0,1]$, which are right continuous and have limits, endowed with Skorokhold topology (Billingsley, 1968); and denote "$\Longrightarrow$" as weak convergence in $\mathcal{D}\left[0,1\right]$, hereafter.

Assumption 5. *Assume* $E\left\{\mathbf{IF}\left(\mathbf{Y}_t; \mathbf{F}^m\right)\right\} = \mathbf{0}$ *and* $N^{-1/2}\sum_{t=1}^{\lfloor sN \rfloor} \mathbf{IF}\left(\mathbf{Y}_t; \mathbf{F}^m\right) \Longrightarrow \mathbf{B}_q\left(s\right)$*, where* $\mathbf{\Delta}$ *is a* $q \times q$ *lower triangular matrix with nonnegative diagonal entries and* $\mathbf{B}_q\left(\cdot\right)$ *is a* q*-dimensional vector of independent Brownian motions. The matrix* $\mathbf{\Sigma}\left(\mathbf{F}^m\right) = \mathbf{\Delta}\mathbf{\Delta}^{\mathrm{T}} = \sum_{k=-\infty}^{\infty} \mathrm{cov}\left\{\mathbf{IF}\left(\mathbf{Y}_0; \mathbf{F}^m\right), \mathbf{IF}\left(\mathbf{Y}_k; \mathbf{F}^m\right)\right\}$ *is positive definite.*

Assumption 5 is similar to Assumption (A3) in Kunsch (1989), except that Assumption (A3) in Kunsch (1989) concerns the entire sample. Assumption 5 is referred to as Assumption 1 in Shao (2010), and Assumption 3.1 in Shao and Zhang (2010). Shao (2010) stated that it is not primitive, and cited Assumption 2.1 in Phillips (1987) as the primitive assumption, which are precisely Assumptions 1-4 imposed earlier.

To ensure the remainder term for the expansion in (21) vanishes asymptotically, we impose the following assumption.

Assumption 6. $\sup_{1 \leq k \leq N} \left|k\mathbf{R}_{1,k}\right| = o_p\left(N^{1/2}\right)$ *and* $\sup_{1 \leq k \leq N} \left|k\mathbf{R}_{1T-k+1,N}\right| = o_p\left(N^{1/2}\right)$.

Assumption 6 is the sample analogue of Assumption (A4) in Kunsch (1989), and is also identical to Assumption 3.2 in Shao and Zhang (2010).

Under the expansion in (23), suppose that Assumption 6 and Lemma 5 hold, it is clear that

$$N^{1/2}\left(\widehat{\theta}_{1,N} - \theta\right) \xrightarrow{d} N\left[\mathbf{0}, \Sigma\left(\mathbf{F}_m\right)\right], \tag{25}$$

as $N \to \infty$, where $\theta = E\left(\widehat{\theta}_{1,N}\right)$.

Clearly, given that $\mathbf{\Sigma}(\mathbf{F}^m)$ is positive definite by construction, we can appeal to the so called "square-root-free Cholesky decomposition", and there exists a unique lower triangular matrix A with 1's along the principal diagonal and a unique diagonal matrix D with positive entries along the principal diagonal such that

$$\mathbf{\Sigma}(\mathbf{F}^m) = \mathrm{ADA}^{\mathrm{T}}. \tag{26}$$

Similarly, the sample counterpart of $\mathbf{\Sigma}(\mathbf{F}^m)$ can be estimated using the standard formula for a covariance matrix, i.e.

$$\widehat{\mathbf{\Sigma}}(\mathbf{F}_m) = (N-1)^{-1} \sum_{k=1}^{N} \left(\widehat{\theta}_{1,k} - \overline{\widehat{\theta}}_N\right) \left(\widehat{\theta}_{1,k} - \overline{\widehat{\theta}}_N\right)^{\mathrm{T}}, \tag{27}$$

and because (27) is a positive definite matrix, it too can be decomposed uniquely in a fashion similar to (26),

$$\widehat{\mathbf{\Sigma}}(\mathbf{F}_m) = \widehat{\mathrm{A}}\widehat{\mathrm{D}}\widehat{\mathrm{A}}^{\mathrm{T}}. \tag{28}$$

Extending the univariate KS_n^R to the multivariate setting setting is not as straightforward, because the adjusted-range LRV is always positive, while in a covariance matrix, only the diagonal elements are necessarily positive. In this case, we propose to gather all the off-diagonal covariances into the diagonal components to estimate the diagonal adjusted-range based covariance matrix, then transform it back so as to estimate the actual adjusted-range based covariance matrix. The procedure takes the following three steps.

1. Obtain $\widehat{\mathrm{A}}$ an invertible matrix using (29), and construct a new series of quantities of interest $\left\{\widehat{\theta}^*_{1,k}\right\}$ such that

$$\widehat{\theta}^*_{1,k} = \widehat{\mathrm{A}}^{-1}\left(\widehat{\theta}_{1,k} - \overline{\widehat{\theta}}_N\right), \tag{29}$$

   for $k \in \{1, \ldots, N\}$, whose covariance is diagonal by construction, i.e. $E\left(\widehat{\theta}^{*\mathrm{T}}_{1,k}\widehat{\theta}^*_{1,k}\right) = \widehat{D}$.

2. Define

$$\mathbf{T}^*_n(k) = kN^{-1/2}\left(\widehat{\theta}^*_{1,k} - \widehat{\theta}^*_{1,N}\right), \tag{30}$$

   which is in analogue of (24), and

$$\widetilde{S}^*_k = k\widehat{\theta}^*_{1,k} - \widehat{\theta}^*_{1,N}. \tag{31}$$

We can deduce the $q \times 1$ adjusted-range self-normalizer

$$\widehat{\mathbf{R}}_n(k) = N^{-1/2} \begin{pmatrix} \max_{1\le t\le N}(s_{1,t}) - \min_{1\le t\le N}(s_{1,t}) \\ \vdots \\ \max_{1\le t\le N}(s_{\iota,t}) - \min_{1\le t\le N}(s_{\iota,t}) \\ \vdots \\ \max_{1\le t\le N}(s_{q,t}) - \min_{1\le t\le N}(s_{q,t}) \end{pmatrix},$$

where $\{s_{\iota,t}\} := \{s_{\iota,1}, \ldots, s_{1,N}\}$, $\iota \in \{1, \ldots, q\}$, is the ιth row of the matrix S, which is a matrix constructed by aligning the column vectors $\widetilde{S}_k^*$ (31) according to the time indices, and forming a $q \times N$ matrix - $\left[\widetilde{S}_1^*, \ldots, \widetilde{S}_N^*\right]$. Thus, we obtain the adjusted-range based estimator for the covariance matrix of $\left\{\widehat{\theta}_{1,k}^*\right\}$, viz. $\operatorname{diag}\left(\widehat{\mathbf{R}}_n(k)\right)$.

3. The adjusted-range based covariance matrix estimator for the original series $\left\{\widehat{\theta}_{1,k}\right\}$ is

$$\widehat{\mathrm{A}}^{\mathrm{T}} \operatorname{diag}\left(\widehat{\mathbf{R}}_n(k)\right) \widehat{\mathrm{A}}. \tag{32}$$

Clearly, this procedure effectively solves the problem of extending the adjusted-range to the multivariate case, the premultiplication by $\widehat{\mathrm{A}}^{-1}$ in (29) is subsequently transformed back in the covariance formula in (32). Note $\widehat{\mathrm{A}}$ is constructed using the sample covariance estimator, i.e. (27); there is no need to use a kernel-based LRV estimator. The sample estimator is preferred for the following reasons. The decomposition can be regarded as auxiliary, it is designed to summarize the non-trivial off-diagonal entries of the sample covariance matrix into the diagonal entries; then after obtaining the adjusted-range based long-run covariance matrix estimator based on adjusted series, we transform it back into the adjusted-range based long-run covariance matrix. Throughout this decomposition and transformation, we relied on the same matrix $\widehat{\mathrm{A}}$, which is invariant and unique for a given $\widehat{\Sigma}(\mathbf{F}_m)$; and $\widehat{\mathrm{A}} \xrightarrow{p} \mathrm{A}$ as $n \to \infty$.

In fact, our proposed procedure is similar to both the structural vector autoregression (SVAR) literature and the vector autoregressive (VAR) prewhitening in Andrews and Monahan (1992). See Remarks 1 and 2 respectively.

Remark 1. *Following Gouriéroux et al. (2017), suppose that*

$$\left(\widehat{\theta}_{1,k} - \overline{\widehat{\theta}}_N\right) = \widehat{\mathrm{A}}\widehat{\theta}_{1,k}^*,$$

where $\overline{\widehat{\theta}}_N = N^{-1}\sum_{k=1}^N \widehat{\theta}_{1,k}$, *and* $\overline{\widehat{\theta}}_N \xrightarrow{p} \theta$ *as* $N \to \infty$. *Here* $\left\{\widehat{\theta}_{1,k} - \overline{\widehat{\theta}}_N\right\}$ *have zero-mean and the matrix* $\widehat{\mathrm{A}}$ *is invertible;* $\widehat{\mathrm{A}}$ *is the "mixing matrix" and* $\widehat{\mathrm{A}}^{-1}$ *is the "demixing matrix". That is to say, we could view the sequence* $\left\{\widehat{\theta}_{1,k} - \overline{\widehat{\theta}}_N\right\}_{k=1,\ldots,N}$ *as linear combinations of* $\left\{\widehat{\theta}_{1,k}^*\right\}_{k=1,\ldots,N}$. *The problem with the so-called independence analysis in Gouriéroux et al. (2017) is to identify* $\widehat{\mathrm{A}}^{-1}$ *and* $\widehat{\theta}_{1,k}^*$ *from the knowledge of* $\left\{\widehat{\theta}_{1,k}\right\}$, *which may not always be feasible. For example, if*

$\left\{\widehat{\theta}^*_{1,k}\right\}$ *is Gaussian, which is likely, for instance, when the parameters of interest are the means of a series of multivariate normally distributed variables, then from the knowledge of* $\left\{\widehat{\theta}_{1,k} - \overline{\widehat{\theta}}_N\right\}$, *we could identify the matrix* $\widehat{\mathrm{A}}\widehat{\mathrm{A}}^{\mathrm{T}}$, *but not matrix* $\widehat{\mathrm{A}}$ *itself. For example, if* $\widehat{\mathrm{A}}^* = \widehat{\mathrm{A}}\mathrm{Q}$ *where* Q *is an orthogonal matrix, we have* $\widehat{\mathrm{A}}^*\widehat{\mathrm{A}}^{*\mathrm{T}} = \widehat{\mathrm{A}}\widehat{\mathrm{A}}^{\mathrm{T}}$, *which causes issues for local and global identification. See Gouriéroux et al. (2017) and references within for methods to deal with such identification issues. Fortunately, our objective is not the same as that in the SVAR literature, which is "to derive the dynamic responses of macro-finance variables to so-called structural shocks" (Gouriéroux et al., 2017), our objective is to generate the adjusted-range covariance matrix. We just need to assume the invertibility of* $\widehat{\mathrm{A}}$, *which is satisfied trivially to construct* $\left\{\widehat{\theta}^*_{1,k}\right\}$, *see (29).*

Remark 2. *Andrews and Monahan (1992) found that autocorrelation affects the accuracy of the HAC long-run covariance estimator, and suggested that a VAR model should be fitted to the original series to moderate the autocorrelation. Suppose that we wish to estimate the long-run covariance matrix for the autocorrelated multivariate series* $\{Z_\tau\}$, *we first need to fit a VAR of the form*

$$Z_t = \phi_1 Z_{\tau-1} + \phi_2 Z_{\tau-2} + \cdots + \phi_p Z_{\tau-p} + \upsilon_t, \tag{33}$$

and obtain the HAC covariance estimate based on $\{\widehat{\upsilon}_t\}$, *denoted by* $\widehat{\mathbf{\Sigma}}_\upsilon$; *finally, the HAC estimate for the original series* $\{Z_\tau\}$ *is*

$$\widehat{\mathbf{\Sigma}} = \left(I - \sum_{\tau=1}^{p} \widehat{\phi}_\tau\right)^{-1} \widehat{\Sigma}_\upsilon \left(I - \sum_{\tau=1}^{p} \widehat{\phi}_\tau\right),$$

where I *denotes an identity matrix of matching dimension. Our "square-root-free Cholesky decomposition" approach is similar to VAR prewhitening, and the similarity is reflected in terms of being able to associate the covariance estimator* $\widehat{\mathbf{\Sigma}}$ *with a sequence of linear regressions as in (33), denote* $\widehat{Z}_\tau$ *as the predictors of* Z_τ *based on* $\{Z_1, \ldots, Z_{\tau-1}\}$ *using linear regressions,* $\upsilon_\tau = Z_\tau - \widehat{Z}_\tau$ *as the residual for such linear regressions, and set* $\upsilon_1 = Z_1$. *Denote the variance of* υ_τ *as* σ^2_τ. *Thus, for* $1 < \tau \leq p$, *there are unique coefficients* $\{\phi_{\tau j}\}$ *such that*

$$Z_\tau = \sum_{j<\tau} \phi_{\tau j} Z_j + \upsilon_\tau. \tag{34}$$

Denote Φ *as the lower triangular matrix with elements* $\phi_{\tau j}$, *where* $1 \leq j < \tau \leq p$. *Clearly, (34) is equivalent to* $(I - \Phi)^{-1}\,\mathbf{Z} = \boldsymbol{\nu}$, *where* $\mathbf{Z} = (Z_1, \ldots, Z_p)^{\mathrm{T}}$, $\boldsymbol{\nu} = \{\nu_1, \ldots, \nu_p\}^{\mathrm{T}}$, *and*

$$\widehat{\mathbf{\Sigma}} = (I - \Phi)^{-1}\, diag\left(\sigma_1^2, \ldots, \sigma_p^2\right) \left[(I - \Phi)^{-1}\right]^{\mathrm{T}}. \tag{35}$$

Therefore, from (28) and (35), it is not difficult to see that $\widehat{\mathrm{A}} = (I - \Phi)^{-1}$ *and* $\widehat{\mathrm{D}} = diag\left(\sigma_1^2, \ldots, \sigma_p^2\right)$. *To summarize, our "square-root-free Cholesky decomposition" approach associates the covariance*

estimator $\widehat{\mathbf{\Sigma}}$ *with a sequence of linear regressions such that*

$$\begin{aligned} Z_1 &= \upsilon_1, \\ Z_\tau &= \mathbf{a}_\tau^{\mathrm{T}} \boldsymbol{\nu} = \sum_{j<\tau} a_{\tau j} \upsilon_j + \upsilon_\tau, \end{aligned} \tag{36}$$

where $\tau = 2, \ldots, p$*, and* $\mathbf{a}_\tau$ *is the* τ*th row of* $\widehat{\mathrm{A}}$*. Clearly, the set of linear regressions in (36) is similar to that of (33).*

Based on the adjusted-range based covariance matrix estimator (29), we obtain the following adjusted-range EKS type of statistics

$$\begin{aligned} \mathrm{EKS}_n^R(q) &= \max_{1 \le k \le N-1} \mathbf{T}_n^*(k)^{\mathrm{T}} \left[\mathrm{diag}\left(\widehat{\mathbf{R}}_n(k) \right) \right]^{-2} \mathbf{T}_n^*(k) \\ &= \max_{1 \le k \le N-1} \mathbf{T}_n(k)^{\mathrm{T}} \left[\widehat{\mathrm{A}}^{\mathrm{T}} \mathrm{diag}\left(\widehat{\mathbf{R}}_n(k) \right) \widehat{\mathrm{A}} \right]^{-2} \mathbf{T}_n(k). \end{aligned} \tag{37}$$

When $q = 1$, $\mathrm{EKS}_n^R(1)$ is equivalent to the square of the simpler KS_n^R. Hence, for the $\mathrm{EKS}_n^R(q)$ statistics, we normally consider $q \ge 2$.

One of the major advantages of $\mathrm{EKS}_n^R(q)$ is the relatively neater formulation in comparison with G_n. Under Lemma 5,

$$\mathrm{diag}\widehat{\mathbf{R}}_n(k) \Rightarrow \mathrm{diag}\left[\sup_{s \in [0,1]} \mathbb{B}_q(s) - \inf_{s \in [0,1]} \mathbb{B}_q(s) \right],$$

where $\mathbb{B}_q(s)$ is the q-dimensional Brownian bridge; and using the CMT, we obtain Theorem 3.

Theorem 3. *Under Lemma 5 and Assumption 6, the null distribution for* $\mathrm{EKS}_n^R(q)$ *is as follows,*

$$\mathrm{EKS}_n^R(q) \xrightarrow{d} \sup_{s \in [0,1]} |\mathbb{B}_q(s)|^{\mathrm{T}} \mathrm{diag}\left[\sup_{s \in [0,1]} \mathbb{B}_q(s) - \inf_{s \in [0,1]} \mathbb{B}_q(s) \right]^{-2} |\mathbb{B}_q(s)|. \tag{38}$$

The proof of Theorem 3 can be found in Appendix A.

Note that we do not necessarily have to conduct the decomposition as in (26). Sometimes there might be implicit/explicit assumptions that the different entries in θ are independent. For instance, the sample mean and variance derived from a normally distributed series are independent of each other. Or if we are testing the constancy of parameters, we may have reasonable grounds to believe that errors or the one-step ahead prediction errors are uncorrelated; see the supplementary material for further detail. In these cases, we can work with

$$\widetilde{S}_k = k\widehat{\theta}_{1,k} - \widehat{\theta}_{1,N}$$

instead of $\widetilde{S}_k^*$, and the EKS statistic becomes

$$\widetilde{\mathrm{EKS}}_n^R(q) = \max_{1\le k\le N-1} \mathbf{T}_n(k)^{\mathrm{T}} \left[\widehat{\mathbf{R}}_n(k)\right]^{-1} \mathbf{T}_n(k)$$
$$\xrightarrow{d} \sup_{s\in[0,1]} |\mathbb{B}_q(s)|^{\mathrm{T}} \operatorname{diag}\left[\sup_{s\in[0,1]} \mathbb{B}_q(s) - \inf_{s\in[0,1]} \mathbb{B}_q(s)\right]^{-2} |\mathbb{B}_q(s)|. \tag{39}$$

Clearly, $\widetilde{\mathrm{EKS}}_n^R(q)$ (39) shares the same asymptotic distribution as $\mathrm{EKS}_n^R(q)$ (37).

Consider a more specific alternative hypothesis, viz.

$$\mathbb{H}_1 : \theta_1 = \cdots = \theta_{k^*} \neq \theta_{k^*+1} = \cdots = \theta_N.$$

Then the test statistic of Shao and Zhang (2010) is

$$\mathrm{G}_n(q) = \max_{1\le k\le N} \mathbf{T}_n(k)^{\mathrm{T}} \left[\mathbf{V}_n(k)\right]^{-1} \mathbf{T}_n(k), \tag{40}$$

and the self-normalizer takes the following form:

$$\mathbf{V}_n(k) = N^{-2}\left\{\sum_{t=1}^{k} t^2 \left(\widehat{\theta}_{1,t} - \widehat{\theta}_{1,k}\right)\left(\widehat{\theta}_{1,t} - \widehat{\theta}_{1,k}\right)^{\mathrm{T}}\right.$$
$$\left. + \sum_{t=k+1}^{N} (N-t+1)^2 \left(\widehat{\theta}_{t,N} - \widehat{\theta}_{k+1,N}\right)\left(\widehat{\theta}_{t,N} - \widehat{\theta}_{k+1,N}\right)^{\mathrm{T}}\right\}.$$

$\mathbf{V}_n(k)$ is an analogue of the forward and backward partial sum as in (13). As a result, and under Lemma 5 and Assumption 6, we have

$$\mathrm{G}_n(q) \xrightarrow{d} \sup_{s\in[0,1]} \mathbb{B}_q(s)^{\mathrm{T}} \mathbf{V}_q(s)^{-1} \mathbb{B}_q(s), \tag{41}$$

where $\mathbf{V}_q(s) = \int_0^s \mathbb{B}_{1,q}(r,s)\,\mathbb{B}_{1,q}(r,s)^{\mathrm{T}}\,dr + \int_s^1 \mathbb{B}_{2,q}(r,s)\,\mathbb{B}_{2,q}(r,s)^{\mathrm{T}}\,dr$, $\mathbb{B}_{1,q}(r,s) = \mathbf{B}_q(r) - (r/s)\,\mathbf{B}_q(s)$ for $r \in [0,s]$, and $\mathbb{B}_{2,q}(r,s) = [\mathbf{B}_q(1) - \mathbf{B}_q(r)] - (1-r)/(1-s)\,[\mathbf{B}_q(1) - \mathbf{B}_q(s)]$ for $r \in [s,1]$.

The result (41) is stated in Theorem 1 of Shao and Zhang (2010), and the critical values for $\mathrm{G}_n(q)$ are tabulated therein as Table 1. The $\mathrm{EKS}_n^R(q)$ statistics have a neater formulation than $\mathrm{G}_n(q)$, as they do not involve the backward and forward summation of a multivariate process. Furthermore, $\mathrm{EKS}_n^R(q)$ is a portmanteau test and may be expected to have power against a broader range of alternatives than $\mathrm{G}_n(q)$, which is formulated under a one-change point alternative.

The asymptotic distribution and the critical values for $\mathrm{EKS}_n^R(q)$ can be approximated using Monte Carlo simulation. Following Shao and Zhang (2010), we set the number of Monte Carlo replications to be $10,000$ and the Brownian motion is approximated using $5,000$ i.i.d. $N(0,1)$ variables. The asymptotic critical values are tabulated in Table 2. The histograms and corresponding

empirical densities are visualized in Figure 3. Clearly, as q gets larger, the distribution becomes more dispersed.

[Insert Table 2 here.]

[Insert Figure 3 here.]

4 The adjusted-range based self-normalized tests for parameter constancy

Here we demonstrate that the adjusted-range KS and EKS statistics can be used to test for parameter constancy. We consider a null model that can be estimated using the ordinary least squares (OLS), if no lagged dependent variables are incorporated, or conditional least square (CLS), when lagged dependent variables are present. Following Klimko and Nelson (1978), we specify the null model such that,

$$E\left(Y_t \middle| \mathfrak{F}_{t-1}\right) = g\left(\theta, \mathfrak{F}_{t-1}\right) \tag{42}$$

where θ is a $p \times 1$ vector, $\mathfrak{F}_{t-1}$ is a sigma-algebra formed by past values of the to be explained Y_t and a vector of exogenous covariates X_t; X_t is $\mathfrak{F}_{t-1}$ measurable and obeys weak laws of large numbers. The one-step ahead prediction error is

$$r_t = Y_t - E\left(Y_t \middle| \mathfrak{F}_{t-1}\right), \tag{43}$$

which is a martingale difference sequence (MDS) with respect to $\mathfrak{F}_{t-1}$, such that $E\left(r_t \middle| \mathfrak{F}_{t-1}\right) = 0$ and $E\left(r_t^2 \middle| \mathfrak{F}_{t-1}\right) = \sigma_r^2$. The null model is general enough to include various types of models, for instance, the linear regression model, the autoregressive (AR) model and some of the nonlinear AR models, such as the the integer valued autoregressive (INAR) model, which is a collection of models that account for serial correlation in count data.

Under non-constancy of parameters, the parameter vector θ becomes time-varying and so does the conditional expectation, $E\left(Y_t \middle| \mathfrak{F}_{t-1}\right)$. We consider the null hypothesis

$$\mathbb{H}_0 : \theta_t = \theta,$$

versus the alternative hypothesis

$$\mathbb{H}_1\text{: otherwise;}$$

when the KS test statistics are entertained. Alternatively, when the G test statistic is used, we consider the case when parameter instability presents as a structural break, i.e.

$$\theta_t = \begin{cases} \theta & \text{for } t = 1, \ldots, n_B \\ \theta + \Delta_\theta & \text{for } t = n_B + 1, \ldots, n \end{cases}, \tag{44}$$

where the break point n_B and the magnitude of shift vector Δ_θ are both unknown. The null hypothesis is

$$\mathbb{H}_0 : \Delta_\theta = 0,$$

and the alternative is

$$\mathbb{H}_1 : \Delta_\theta \neq 0.$$

Following Klimko and Nelson (1978), we impose the following conditions for OLS/CLS to be asymptotically normal and consistent.

Assumption 7. *The sequences $\{X_t\}$ and $\{Y_t\}$ are stationary and ergodic with finite 4th moments.*

Assumption 8. *The function $g(\theta|\mathfrak{F}_{t-1})$ and its partial derivatives are continuous and uniformly bounded in θ for all i, j, k, such that*

$$\begin{aligned} |g| &\leq H^0, \\ |dg/d\theta_i| &\leq H_i^1, \\ \left|d^2g/d\theta_i d\theta_j\right| &\leq H_{ij}^2, \\ \left|d^3g/d\theta_i d\theta_j d\theta_k\right| &\leq H_{ijk}^3, \end{aligned}$$

where the H are square integrable functions, which are independent of θ, but may depend on $\{X_t\}$ and $\{Y_t\}$.

Assumption 9. *The matrix defined by $E[dg/d\theta_i.dg/d\theta_j]$ is nonsingular and $dg(\theta, \mathfrak{F}_{t-1})/d\theta_i$ has finite 4th moments.*

Assumptions 7-9 are sufficient conditions for the OLS/CLS estimator to be asymptotically normal and consistent. Assumption 7 is in analogue to the regularity condition of the influence function as in Assumption 1. It is not difficult to see that the consistency and asymptotic normality of $\widehat{\theta}$ in (45) relies on the Taylor expansion of function $g(\cdot)$, that is why Assumptions 8-9 address the regularity conditions of its derivatives with respect to θ.

Under Assumptions 7-9, the parameter estimates for the null model are obtained through minimizing the sum of squared deviations with respect to θ,

$$\widehat{\theta} = \arg\min_{\theta} \sum_{t=1}^{n} (r_t)^2, \tag{45}$$

and $\widehat{\theta}$ is consistent and asymptotically normal. The estimated one-step ahead prediction error is

$$\widehat{r}_t = Y_t - g\left(\widehat{\theta}, \mathfrak{F}_{t-1}\right).$$

The test for parameter constancy is equivalent to testing structural breaks in the mean of estimated one-step ahead prediction errors.

Consequently, we can evaluate KS and G test statistics under $\{\widehat{r}_t\}$, and obtain the following constancy of parameter tests, viz.

$$\mathrm{KS}_{r,n}^{0} = \max_{1\leq j\leq n} n^{-1/2}\sum_{t=1}^{n}\widehat{\sigma}_r^{-1}\left|\sum_{t=1}^{j}\left(\widehat{r}_t-\overline{\widehat{r}}\right)\right| \xrightarrow{d} \sup_{s\in[0,1]}\left|\mathbb{B}\left(s\right)\right|,$$

$$\mathrm{KS}_{r,n}^{(b)} = \max_{1\leq j\leq n} n^{-1/2}\widehat{V}_r\left(b\right)^{-1/2}\left|\sum_{t=1}^{j}\left(\widehat{r}_t-\overline{\widehat{r}}\right)\right| \xrightarrow{d} \frac{\sup_{s\in[0,1]}\left|\mathbb{B}\left(s\right)\right|}{\sqrt{Q\left(b\right)}},$$

$$\mathrm{KS}_{r,n}^{W} = \max_{1\leq j\leq n} n^{-1}W_{r,n}^{-1}\left|\sum_{t=1}^{j}\left(\widehat{r}_t-\overline{\widehat{r}}\right)\right| \xrightarrow{d} \frac{\sup_{s\in[0,1]}\left|\mathbb{B}\left(s\right)\right|}{\sqrt{\int_0^1\left[\mathbb{B}\left(s\right)\right]^2 ds}},$$

$$\mathrm{G}_{r,n} = \max_{1\leq k\leq n-1}\mathrm{T}_{r,n}\left(k\right)'\left[V_{r,n}\left(k\right)\right]^{-1}\mathrm{T}_{r,n}\left(k\right) \xrightarrow{d} \sup_{s\in[0,1]}\mathbb{B}\left(s\right)'V^{-1}\left(s\right)\mathbb{B}\left(s\right),$$

and

$$\mathrm{KS}_{r,n}^{R} = \max_{1\leq j\leq n} n^{-1/2}\widehat{R}_{r,n}^{-1}\left|\sum_{t=1}^{j}\left(\widehat{r}_t-\overline{\widehat{r}}\right)\right| \frac{\sup_{s\in[0,1]}\left|\mathbb{B}\left(s\right)\right|}{\sup_{s\in[0,1]}\mathbb{B}\left(s\right)-\inf_{s\in[0,1]}\mathbb{B}\left(s\right)},$$

where $\overline{\widehat{r}} = n^{-1}\sum_{t=1}^{n}\widehat{r}_t$, and $\mathrm{T}_{r,n}\left(k\right)$, $\widehat{V}_r\left(b\right)$, $W_{r,n}^{-1}$, $V_{r,n}\left(k\right)$ and $\widehat{R}_{r,n}$ are simply (6), (7), (8), (11) and (16) are evaluated at $\{\widehat{r}_t\}$.

The asymptotic results follow easily under the assumptions used. Assumption 1 is satisfied trivially given $\{r_t\}$ is the one-step ahead prediction error, which together with Assumptions 2-4, would ensure that KS_n^R, KS_n^W, KS_n^0 and $\mathrm{KS}_n^{(b)}$ and G_n evaluated under $\{r_t\}$ follow the same asymptotic distributions. Because the parameter estimators are asymptotically consistent and normal (under Assumptions 7-9), the remainder terms for a Taylor expansion of the statistics evaluated around the real θ are negligible (Han and McCabe, 2013). Thus, it is not difficult to deduce that $\mathrm{KS}_{r,n}^R$, $\mathrm{KS}_{r,n}^W$, $\mathrm{KS}_{r,n}^0$, $\mathrm{KS}_{r,n}^{(b)}$ and $\mathrm{G}_{r,n}$ also follow the same distributions.[3]

Similarly, if the test statistics $\mathrm{EKS}_n^R\left(q\right)$ (37), $\widetilde{\mathrm{EKS}}_n^R\left(q\right)$ (39) and $\mathrm{G}_n\left(q\right)$ (40) are evaluated at the multivariate estimated one-step ahead prediction error $\widehat{\mathbf{r}}_t$, instead of $\mathbf{Y}_t$, they become multivariate tests for parameter constancy, denoted by $\mathrm{EKS}_{r,n}^R\left(q\right)$, $\widetilde{\mathrm{EKS}}_{r,n}^R\left(q\right)$ and $\mathrm{G}_{r,n}\left(q\right)$ respectively. The asymptotic distributions remain the same and so does the rationale behind the proofs as used in the univariate case.

[3]Note that Assumption 1 requires the one-step ahead prediction errors to be strong mixing. By construction the series of one-stead ahead prediction errors is a MDS. While MDS doesn't necessarily imply strong mixing, it is still reasonable to assume that an MDS sequence is a strong mixing process. For instance, studies such as Hall and Heyde (1980) approximated mixing sequences by martingales, see pp. 19-20 in Hall and Heyde (1980) for examples of martingales that are mixingales. Moreover, McLeish (1975) introduced the concept of mixingales as asymptotic martingales, and observed that mixingales are sufficiently like MDS to satisfy convergence theorems.

5 Simulation studies

Here we consider structural break in the mean for a shift in the level of an AR process (DGP1), an increase in the persistence of autocorrelation in a univariate series (DGP2), and a level shift in a bivariate VAR(1) model (DGP3). Our focus on the mean is motivated by the equivalence of testing structural changes in other quantities to testing the mean of an alternative sequence (Carlstein et al., 1994; Brodsky and Darkhovsky, 2005; Harchaoui and Lévy-Leduc, 2010). Throughout the simulation studies, we set the number of Monte Carlo replications to be 1000, the sample sizes to be 500 and 1000 for univariate cases (DGP1 and DGP2), and 250 and 500 for multivariate case (DGP3). The significant level used is 5%.

5.1 DGP1 - shift in the level in an AR model

Here we consider a non-negative shift in the level of an linear AR model, similar to Shao and Zhang (2010). Although we modify the DGP in Shao and Zhang (2010) to allow an integrated GARCH error process, this arrangement reflects the widely accepted norm that there are phases of nonstationarity in financial time series (Shephard, 2010). Moreover, the widely applied RiskMetrics developed by J. P. Morgan (1997) is a special case of the integrated GARCH(1,1) model. This specification also follows the simulation studies of Mandelbrot (1972) and Mandelbrot (1975), which showed the almost-sure convergence of the range statistic for stochastic processes with infinite variances. Finally, this setup reflects the focus of modelling financial volatility using range statistics (Chou et al., 2010).

Let

$$u_t = 0.5u_{t-1} + \epsilon_t, \tag{46}$$

and ϵ_t follows a simple integrated GARCH process, such that

$$\begin{aligned} \epsilon_t &= h_t \varepsilon_t, \\ h_t^2 &= 0.01 + \epsilon_{t-1}^2, \end{aligned} \tag{47}$$

where (46) and (47) are the mean equation and the variance equation respectively, with $\varepsilon_t \sim$ i.i.d. $N(0,1)$ and $t = 1, \ldots, n$. Set

$$y_t = \begin{cases} u_t, \ 1 \leq t \leq n/2 \\ \eta + u_t, \ n/2 + 1 \leq t \leq n \end{cases}, \tag{48}$$

where $\eta \geq 0$, i.e. we consider a non-negative shift of level. We consider $\eta \in \{0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$. Hence, the time series $\{y_t\}$ follows an autoregressive process under the null ($\eta = 0$), and under the alternative, there is a level shift of magnitude η in the middle of the sample. Other than powers, we also report the size-adjusted-powers, which are the powers generated based on empirical critical values. The sizes, powers and size-adjusted-powers are tabulated in Table 3. Note because $\{y_t\}$ is univariate, we consider the KS_n^R statistic.

[Insert Table 3 here.]

From the simulation results, it is clear that the G_n and KS_n^R test statistics are the most reliable. However, KS_n^R suffers from over inflation of size when the sample size is small, but this rectifies itself as the sample size gets larger, or when the dimension increases, see Section 5.3. Also, the over-inflation of size is less severe if we assume the error terms are stationary, e.g. the DGP considered in Shao and Zhang (2010), or even if we include outliers in the error terms. These simulations are omitted for brevity, but are available from the authors upon request.

The cause of such over inflation of sizes for KS_n^R is that the minimum and maximum statistics converge slowly to their asymptotic values. In particular, Hall (1979) studied the largest order statistic and found that the rate of convergence of the maximum to the extreme value distribution is slow. See Welsh (2018) for coverage of Peter Hall's early research, which was heavily concentrated on the theme of extreme value statistics.

The KS statistics based on the naive self-normalization, small-b and fixed-b asymptotics are also inadequate compared to G_n and KS_n^R. The percentage of rejections for KS_n^0 first increases and then declines; clearly, KS_n^0 suffers from the nonmonotonic power problem. The KS_n^W and $\mathrm{KS}_n^{(b)}$ statistics do not have adequate power. Interestingly, for small $b = 0.1$, the pattern of rejection percentages of $\mathrm{KS}_n^{(b)}$ is similar to that of $\mathrm{KS}_n^{(0)}$; on the other hand, when b increases, the pattern bears more resemblance to that of $\mathrm{KS}_n^{(W)}$. This is explained in terms of the self-normalization based on Lobato (2001) and Shao (2010) being equivalent to the fixed-b approach when $b = 1$.

5.2 DGP2 - increase in persistence of autocorrelation in an AR model

Here we consider the case when there are shifts in the AR coefficient and, in addition, we set the probability density function (PDF) of the error terms to be a mixture, i.e. $f(\varepsilon) = \sum_{i=1}^{m} w_i p_i(\varepsilon)$, where $m = 3$, $\mathbf{w} = (w_1, w_2, w_3) = (0.4, 0.2, 0.4)$; p_1, p_2, and p_3 are normal PDFs with $\left(\mu^{(1)}, \sigma^{(1)}\right) = (-7.8, 8)$, $\left(\mu^{(2)}, \sigma^{(2)}\right) = (15.2, 3)$, and $\left(\mu^{(3)}, \sigma^{(3)}\right) = (0.2, 6)$. Clearly, $\mu = \sum_{i=1}^{n} w_i \mu^{(i)} = 0$, and the distribution of the error terms is nonstandard with two modes.

We consider the following DGP:

$$y_t = \begin{cases} \alpha_0 y_{t-1} + \varepsilon_t,\ 1 \le t \le n/2 \\ (\alpha_0 + \alpha_1)\, y_{t-1} + \varepsilon_t,\ n/2 + 1 \le t \le n \end{cases}, \tag{49}$$

where $\alpha_0 = 0.1$, and there are 10 possible values for α_1, viz. $\alpha_1 \in \{0.0, 0.30, 0.50, 0.70, 0.90\}$. Unlike (17), where the level of shift η is imposed in the middle of the series, here α_1 is incorporated as an inflation of the parameter, whose effect gets compounded beyond $n/2$. Technically, under the alternative, the structural break presents itself as a smooth change, which is not nested within that of G_n, when only one structural break is allowed. Though, we still compare the performance of G_n, by pretending that we do not know the type of structural break. The sizes, powers and size-adjusted-powers are summarized in Table 4.

[Insert Table 4 here.]

Both KS_n^R and G_n are adequate with increasing powers and size-adjusted-powers as α_1 increases, although KS_n^R again suffers from slight over-inflation of size when the sample size is moderate. However, in addition to its neater formulation, KS_n^R can compete with G_n through having greater powers and size-adjusted-powers. This finding confirms that the adjusted range of a partial sum is particularly suitable for use in series with persistent positive autocorrelation.

It is also clear that KS_n^W is not able to detect the shifts in the AR coefficients; the percentage of rejections is small and the powers and size-adjusted-powers first increase as α_1 increases and then decrease, which is similar to the case of the level shift considered by Shao and Zhang (2010). Furthermore, KS_n^0 suffers from the same problem. Our results reflect the finding that kernel-based LRV estimators under small-b asymptotics tend to provide unreliable estimates in finite samples with only a moderate level of dependence (Müller, 2007).

The $\mathrm{KS}_n^{(b)}$'s are not appropriate for detecting large increases in persistence in autocorrelations, as they have almost no power, with the percentage of rejections below the significance level and the size-adjusted-powers well below that of KS_n^R and G_n. This finding supplements the argument of Shao (2015), viz. the size of the fixed-b based HAC tests deteriorates when the series is a unit-root or close to a unit-root process.

5.3 DGP3 - a level shift in a bivariate VAR(1) model

Here, we consider a level shift in a bivariate VAR(1) model. Let $\mathbf{u}_t = \boldsymbol{\theta}\boldsymbol{u}_{t-1} + \boldsymbol{\varepsilon}_t$, where $\boldsymbol{\varepsilon}_t$ is assumed to follow an i.i.d. multivariate normal distribution with zero mean and a 2×2 dimensional covariance matrix $\mathrm{cov}\,(\boldsymbol{\varepsilon}_t)$. Let

$$\mathbf{y}_t = \begin{cases} \mathbf{u}_t,\ 1 \le t \le n/2 \\ \boldsymbol{\eta} + \mathbf{u}_t,\ n/2+1 \le t \le n \end{cases}, \tag{50}$$

and we consider different values for $\boldsymbol{\eta}$, viz. $\boldsymbol{\eta} \in \left\{(0.0, 0.0)^{\mathrm{T}}, (0.5, 0.5)^{\mathrm{T}}, \ldots, (5.0, 5.0)^{\mathrm{T}}\right\}$. So $\boldsymbol{\eta} = (0.0, 0.0)^{\mathrm{T}}$ corresponds to the null, whereas $\boldsymbol{\eta} = (0.5, 0.5)^{\mathrm{T}}$, $(1.0, 1.0)^{\mathrm{T}}$, ..., and $(5.0, 5.0)^{\mathrm{T}}$ indicate the alternatives. Clearly, under the alternatives there is a level shift in the middle of the series similar to the DGP considered by Shao and Zhang (2010) and Section 5.1.

We consider the following two cases.

(i). $\boldsymbol{\theta} = \begin{pmatrix} 0.5 & 0.0 \\ 0.0 & 0.5 \end{pmatrix}$ and $\mathrm{cov}\,(\boldsymbol{\varepsilon}_t) = \begin{pmatrix} 1.0 & 0.0 \\ 0.0 & 1.0 \end{pmatrix}$,

(ii). $\boldsymbol{\theta} = \begin{pmatrix} 0.5 & 0.1 \\ 0.1 & 0.5 \end{pmatrix}$ and $\mathrm{cov}\,(\boldsymbol{\varepsilon}_t) = \begin{pmatrix} 1.0 & 0.1 \\ 0.1 & 1.0 \end{pmatrix}$.

The first case is identical to two independent series generated from a shift in the level in a univariate linear AR process considered by Shao and Zhang (2010), being aligned together to

form a bivariate series, whereas the second case allows some temporal dependence in $\mathbf{u}_t$ and cross dependence in the error terms $\boldsymbol{\varepsilon}_t$. We considered sizes, powers and adjusted-powers for $\mathrm{EKS}_n^R(2)$ and $\mathrm{G}_n(2)$. The results are summarized in Table 5.

[Insert Table 5 here.]

Despite being able to rectify the nonmonotonic power issue when the series is univariate, when we consider bivariate series the powers of $\mathrm{G}_n(2)$ become inadequate. This reflects the fact that $\mathrm{G}_n(2)$ is still formulated under self-normalization based on the variance of the partial sum process. As the dimension increases, the self-normalizer $\mathbf{V}_n(k)$ (41) quickly increases, just like the increase of W_n in the KS_n^W test. The effect of the backward and forward summation for accommodating the increase in W_n in the univariate case, is clearly moderated/diminished by the increase in $\mathbf{V}_n(k)$ as the dimension increases. In contrast to $\mathrm{G}_n(2)$, $\mathrm{EKS}_n^R(2)$ has increasing power as the magnitude of shift increases, and the results for powers and size-adjusted-powers are not hugely affected by the incorporation of temporal dependence in $\mathbf{u}_t$ and cross dependence in $\boldsymbol{\varepsilon}_t$. Overall, these findings are consistent with those of Wang and Shao (2020), i.e. a self-normalized statistic based on the variance of a partial sum has "better size but less power". Furthermore, they confirm the robustness properties of the range as suggested by the studies of Mandelbrot, Taqqu and Wallis; see Section 1.

6 Empirical analysis

Motivated by the fact that range has been widely applied in volatility estimation (Chou et al., 2010), here we consider changes in conditional heteroskedasticity in ten of the world's major stock indices over the recent past, i.e. from 1st January 2012 to 31st December 2020. The stock indices considered are the Dow Jones Industrial Average, the NYSE Composite, the S&P Composite Index, the FTSE 100, the CAC 40, the FTSE MIB, the Shanghai Stock Composite, the Nikkei 225 and the KOSPI 200, which covers the United States, Canada, the United Kingdom, the European Union (France, Germany and Italy), China, Japan and South Korea. Not all markets are open on the same days, but too many observations would be lost if we were to remove all the days when there were no observations. Thus, we only remove weekend days, then we use the R command "`na.interp`" in the "`forecast`" package to interpolate the indices, and compute the logarithmic return series. There are in total 2385 observations in each return series $\left\{r_t^{(j)}\right\}$, $1 \leq t \leq 2385$ and $1 \leq j \leq 10$, although for notational simplicity, the superscript j is dropped hereafter. The descriptive statistics for daily returns are summarized in Table 6. The series appear stationary, however they all suffer from a high level of kurtosis and non-normality, which are well known characteristics of time series of financial returns.

[Insert Table 6 here.]

6.1 Structural breaks tests in the mean of conditional heteroskedasticity

Here we test whether there are structural breaks in the mean of conditional heteroskedasticity, first individual series, then in a multivariate context. We use the rolling window method, and set the length of the window to be 500, which roughly equals the number of trading days in two years. The step is set to be 1, and there are 1886 steps in total.

First, we investigate if there are structural breaks in the mean of the stochastic volatilities in the individual stock markets. For each individual return series, we fit an autoregressive model,

$$r_t = c_0 + \sum_{i=1}^{p} a_i r_{t-i} + \epsilon_t,$$

where the lag order p is selected using the AIC. Then the fitted residuals are squared to approximate the conditional heteroskedasticity, (i.e. $\{\widehat{\epsilon}_t^2\}$); and these are used to test whether there are structural breaks in the mean of the stochastic volatilities in each of the major stock markets. Note that the significance level is set to be 5%. The results are summarized in Table 7.

[Insert Table 7 here.]

The findings are as follows. For individual series of rates of return, the percentages of rejection are very small and sometimes even zero for the KS statistics based on naive self-normalization (KS_n^W), small-b (KS_n^0) and fixed-b ($\mathrm{KS}_n^{(b)}$) asymptotics. However, the results based on KS_n^W, KS_n^0 and $\mathrm{KS}_n^{(b)}$ should not be used in practice, given their lack of power reflected from the simulation studies in Section 5. Meanwhile, there is significant evidence that there are recurrent structural breaks in conditional volatilities, as the rejection rates for both KS_n^R and G_n are well above the 5% significance level. This may be expected, as generally a stock market would transit through bullish and bearish phrases, with bullish phrases being characterized with low levels of volatilities, and bearish phases otherwise. Clearly, KS_n^0, $\mathrm{KS}_n^{(b)}$ and $\mathrm{KS}_{r,n}^W$ appear to be unsuitable for use in practice due to lack of power, which echoes the findings from the extensive simulation studies in Section 5.

Second, we consider structural breaks in conditional heteroskedasticity in the multivariate rates of return. We fit a vector autoregressive model to the multivariate return series $\{\mathbf{r}_t\}$, $\mathbf{r}_t = \mathbf{c}_0 + \sum_{i=1}^{P} \mathbf{a}_i \mathbf{r}_{t-i} + \boldsymbol{\epsilon}_t$, using the `R` package '`vars`', the order P is selected using the Akaike information criterion (AIC). We obtain the squared residuals $\widehat{\boldsymbol{\epsilon}}_t^2$. Then, we conduct the $\mathrm{G}_n(10)$ and $\mathrm{EKS}_n^R(10)$ tests on $\widehat{\boldsymbol{\epsilon}}_t^2$. The average values for $\mathrm{G}_n(10)$ and $\mathrm{EKS}_n^R(10)$ are 44.6745 and 563.5103, while the percentage of rejections are 97.04% and 100.00% respectively, indicating appreciable evidence that there are significant level shifts in the volatility process.

6.2 Constancy of parameter tests

Here, we consider the suitability of the univariate and multivariate autoregressive conditional heteroskedasticity (ARCH) family of models to describe the evolution of conditional heteroskedasticity

for the return series. For the univariate case, given that an ARCH model is equivalent to an infinite order generalized autoregressive conditional heteroskedasticity (GARCH) model in theory, rejection of parameter constancy for the best fitted ARCH model would suggest that a GARCH may be unsuitable as well.

Consider an AR(P)-ARCH(p) specification as follows,

$$r_t = c + \sum_{i=1}^{P} a_i r_{t-i} + \epsilon_t, \tag{51}$$

$$\epsilon_t = h_t \varepsilon_t,$$

$$h_t^2 = \omega + \sum_{j=1}^{p} \beta_j^2 \epsilon_{t-j}^2, \tag{52}$$

where (51) and (52) are referred to as the mean and variance equations respectively. The standardized innovation $\{\varepsilon_t\}$ is usually assumed to be i.i.d. $N(0,1)$ or sometimes, a MDS$(0,1)$, in order to accommodate the serial dependence in third order and other higher order conditional moments. Clearly, (52) indicates that

$$\epsilon_t^2 = \omega + \sum_{j=1}^{p} \beta_j^2 \epsilon_{t-j}^2 + \xi_t,$$

where $\{u_t\}$ is a MDS. In another word, $\left\{\epsilon_t^2\right\}$ ought to follow an AR process, if the ARCH model can sufficiently model the conditional heteroskedasticity. Thus, we run an AR regression on $\left\{\widehat{\epsilon}_t^2\right\}$ with the order being selected by AIC, and test the new series of residuals $\left\{\widehat{\xi}_t\right\}$. This is equivalent to testing the suitability of an ARCH family model. Note that, as in the manuscript, the significance level is set at 5%. The results are summarized in Table 8.

[Insert Table 8 here.]

Clearly, the ARCH, or more generally speaking, the ARCH family of models is inadequate, given that the percentages of rejections for $\mathrm{KS}_{r,n}^{R}$ and $\mathrm{G}_{r,n}$ are well above the significance level. Again the KS statistics based on naive self-normalization ($\mathrm{KS}_{r,n}^{W}$) or fixed-b asymptotics ($\mathrm{KS}_{r,n}^{b}$) are inadequate due to insufficient power. Clearly, $\mathrm{KS}_{r,n}^{0}$, $\mathrm{KS}_{r,n}^{(b)}$ and $\mathrm{KS}_{r,n}^{W}$ ought not to be applied in practice due to lack of power, which echoes the findings from the extensive simulation studies in Section 5.

For the multivariate case, we first fit a VAR model to the multivariate return series $\{\mathbf{r}_t\}$,

$$\mathbf{r}_t = \mathbf{c}_0 + \sum_{i=1}^{P} \mathbf{a}_i \mathbf{r}_{t-i} + \boldsymbol{\epsilon}_t,$$

then fit a VAR model to $\widehat{\boldsymbol{\epsilon}}_t^2$,

$$\widehat{\boldsymbol{\epsilon}}_t^2 = c^* + \sum_{i=1}^{P^*} \mathbf{a}_i^* \widehat{\boldsymbol{\epsilon}}_{t-i}^2 + \boldsymbol{\xi}_t$$

and calculate $\mathrm{G}_{r,n}(10)$ and $\mathrm{EKS}^R_{r.n}(10)$ on $\widehat{\boldsymbol{\xi}}_t^2$ to examine the suitability of a multivariate ARCH family model. The average values for $\mathrm{G}_{r,n}(10)$ and $\mathrm{EKS}^R_{r.n}(10)$ are 0.2938 and 31.3809, while the percentages of rejection are 1.22% and 0.11%, respectively. These findings would suggest that a multivariate ARCH model can effectively accommodate conditional volatility in all 10 stock markets. Thus, the evidence of unsuitability of the ARCH model in modelling individual series is due to failure to consider volatility spill-overs across different markets.

7 Conclusion

In this paper, we propose using the adjusted sample range of a partial sum as a self-normalizer instead of its variance. Three scenarios are considered: first, testing for structural change in the mean of a series, second, testing for structural changes in a general framework, and third, testing parameter constancy in a regression context.

Similar to the G statistic in Shao and Zhang (2010), our proposed adjusted-range based KS and EKS statistics do not involve any user specified or tuning parameters. There is also no need for forward and backward summations as in the G statistic in Shao and Zhang (2010), or the construction of the contrast process as in Zhang and Lavitas (2018), which simplifies the computation of the test statistics. Moreover, our proposed adjusted-range based KS and EKS test statistics are portmanteau tests, which are robust to multiple structural breaks, whereas the G statistic requires the knowledge of the number of breaks.

The Monte Carlo simulations show that the use of adjusted-range as a self-normalizer can rectify the nonmonotonic power problem present under "naive self-normalization" (Shao and Zhang, 2010). Our proposed KS type statistics offer both adequate sizes and good powers in contrast to the G test statistic in multivariate case, which suffers from the "better size but less power" phenomenon (Wang and Shao, 2020). The simulations on the univariate series in the supplementary material suggest that KS^R_n performs adequately when the errors have an infinite variance and under persistent autocorrelation. The empirical studies on the world's major stock indices also demonstrate the adequacy of the proposed method.

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A Proof for Theorem 3

Following (23), for $1 \leq t_1 \leq t_2 \leq N$, we have

$$t_1 \left(\widehat{\theta}_{1,t_1} - \widehat{\theta}_{1,t_2} \right) = \left\{ \sum_{t=1}^{t_1} \mathbf{IF} \left(\mathbf{Y}_t ; \mathbf{F}^m \right) - \frac{t_1}{t_2} \mathbf{IF} \left(\mathbf{Y}_t ; \mathbf{F}^m \right) \right\} + \left\{ t_1 \mathbf{R}_{1,t_1} - \frac{t_1}{t_2} \mathbf{R}_{1,t_2} \right\}. \tag{53}$$

Under Assumption 6, we can show that the terms in second curly bracket in (53) are negligible, which combined with Lemma 5, the unique decomposition of $\widehat{\Sigma} \left(\mathbf{F}_m \right)$ in (28), viz. $\widehat{\Sigma} \left(\mathbf{F}_m \right) = \widehat{\mathrm{A}} \widehat{\mathrm{D}} \widehat{\mathrm{A}}^{\mathrm{T}}$, and the fact that $\widehat{\mathrm{A}} \xrightarrow{p} \mathrm{A}$ as $N \to \infty$, would imply that the joint convergence of

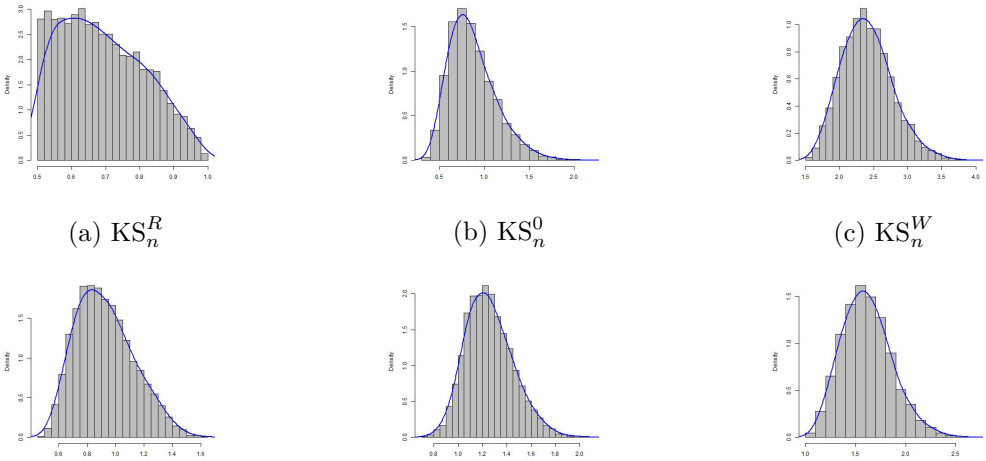
$$\mathbf{T}_n \left(k \right)^{\mathrm{T}} \Rightarrow \boldsymbol{\Delta} \mathbb{B}_q \left(s \right)$$

and

$$\widehat{\mathrm{A}}^{\mathrm{T}} \widehat{\mathbf{R}}_n \left(k \right) \widehat{\mathrm{A}} \Rightarrow \mathrm{diag} \left[\sup_{s \in [0,1]} \mathbb{B} \left(s \right) - \inf_{s \in [0,1]} \mathbb{B} \left(s \right) \right].$$

The result in Theorem 3 is a direct consequence of the CMT. □

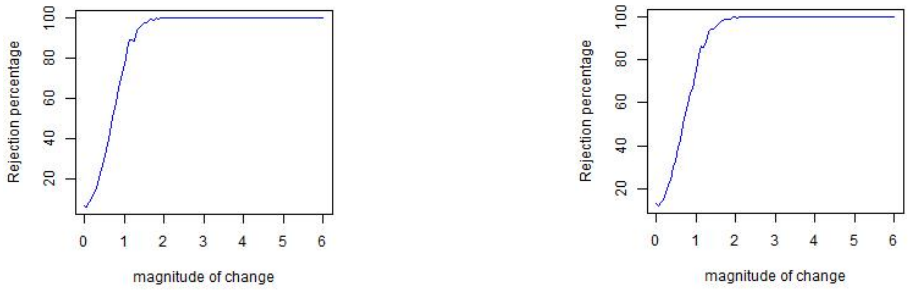
B Figure



(a) KS_n^R (b) KS_n^0 (c) KS_n^W

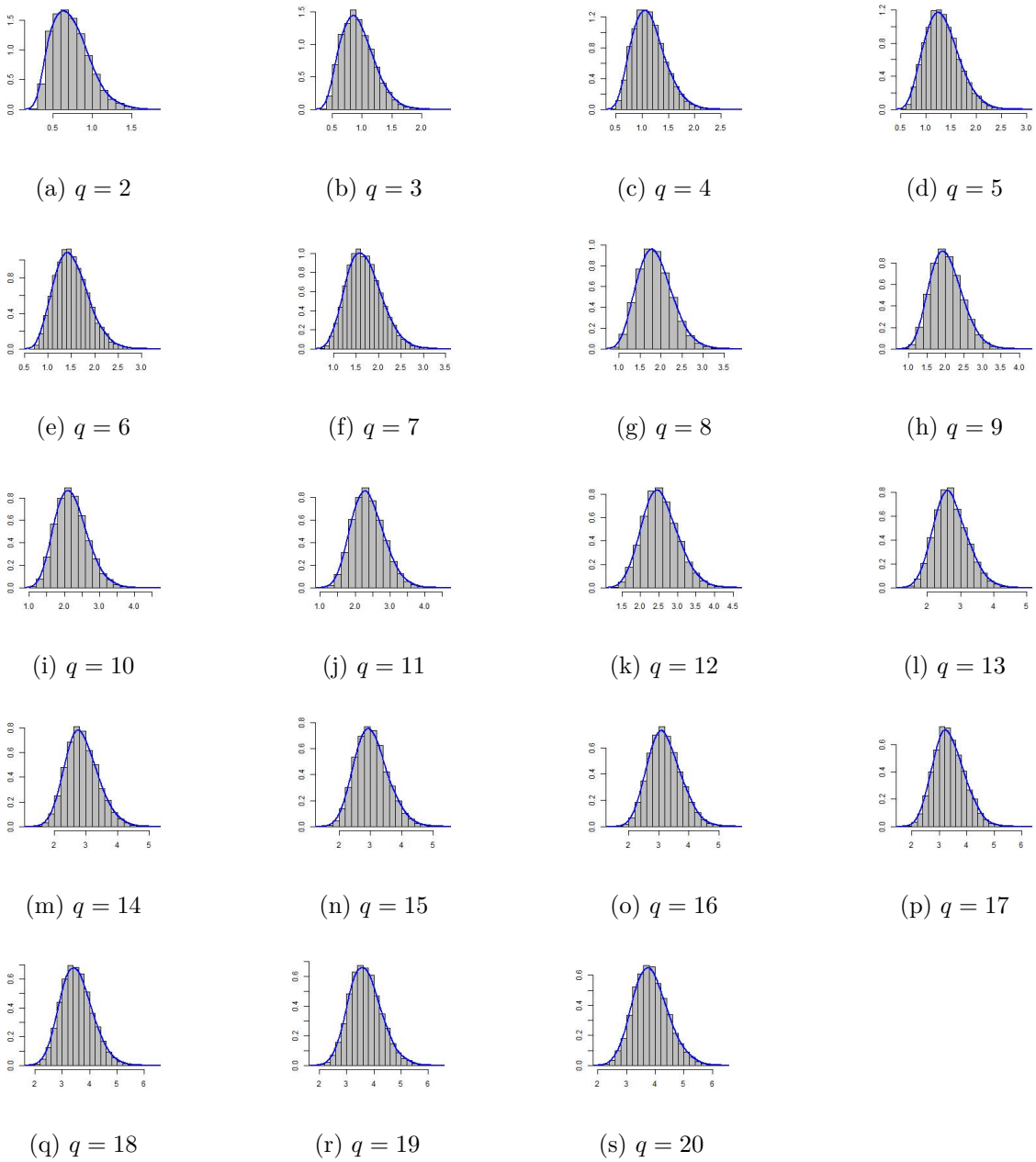
(d) KS_n^b $(b = 0.1)$ (e) KS_n^b $(b = 0.5)$ (f) KS_n^b $(b = 0.9)$

Figure 1: Histograms for one-dimensional KS test statistics and corresponding empirical density functions based on $n = 200,000$ and $10,000$ Monte Carlo replications.



(a) G_n (b) KS_n^R

Figure 2: The rejection percentages of G_n and KS_n^R statistics under shifts in η (17) for $n = 200$.



(a) $q = 2$ (b) $q = 3$ (c) $q = 4$ (d) $q = 5$

(e) $q = 6$ (f) $q = 7$ (g) $q = 8$ (h) $q = 9$

(i) $q = 10$ (j) $q = 11$ (k) $q = 12$ (l) $q = 13$

(m) $q = 14$ (n) $q = 15$ (o) $q = 16$ (p) $q = 17$

(q) $q = 18$ (r) $q = 19$ (s) $q = 20$

Figure 3: Histograms for $\mathrm{EKS}_n^R(q)$ for $q = 2, 3, \ldots, 20$ and corresponding empirical density functions based on $5,000$ i.i.d. $N(0,1)$ variables and $10,000$ Monte Carlo replications.

C Table

Table 1: Simulated critical values for KS test statistics based on $250,000$ i.i.d. $N(0,1)$ and $10,000$ Monte Carlo replications.

Significance level	KS_n^R	KS_n^0	KS_n^W	$\mathrm{KS}_n^{(b)}$		
				(b=0.1)	(b=0.5)	(b=0.9)
10.0%	0.8684	1.2220	2.8857	1.2125	1.5206	1.9350
5.0%	0.9117	1.3640	3.0585	1.2935	1.6099	2.0520
2.5%	0.9391	1.4762	3.2029	1.3552	1.6809	2.1539
1.0%	0.9634	1.6175	3.3896	1.4265	1.7659	2.2698
0.5%	0.9732	1.7119	3.4765	1.4687	1.8215	2.3361
0.1%	0.9869	1.9111	3.6735	1.5652	1.9721	2.5003

Table 2: Simulated critical values for $\mathrm{EKS}_n^R(q)$ for $q = 2, 3, \ldots, 20$ based on $5,000$ i.i.d. $N(0,1)$ and $10,000$ Monte Carlo replications.

	Significance level					
	10.0%	5.0%	2.5%	1.0%	0.5%	0.1%
$q = 2$	1.0339	1.1425	1.2518	1.3706	1.4530	1.5732
$q = 3$	1.2954	1.4216	1.5281	1.6720	1.7793	1.9893
$q = 4$	1.5456	1.6818	1.8003	1.9645	2.0482	2.2967
$q = 5$	1.7692	1.9149	2.0505	2.1939	2.3109	2.7045
$q = 6$	1.9829	2.1544	2.2870	2.4742	2.5882	2.9517
$q = 7$	2.1970	2.3614	2.5163	2.6841	2.8356	3.1257
$q = 8$	2.3971	2.5733	2.7497	2.9263	3.0924	3.3959
$q = 9$	2.6046	2.7860	2.9433	3.1438	3.2838	3.6567
$q = 10$	2.8039	2.9928	3.1787	3.3765	3.5211	3.7879
$q = 11$	2.9760	3.1715	3.3515	3.5603	3.6912	4.1093
$q = 12$	3.1716	3.3744	3.5563	3.7659	3.9302	4.2577
$q = 13$	3.3709	3.5771	3.7660	3.9903	4.1375	4.4623
$q = 14$	3.5513	3.7740	3.9712	4.1957	4.3488	4.6842
$q = 15$	3.7345	3.9558	4.1637	4.4111	4.6053	5.0097
$q = 16$	3.9228	4.1404	4.3300	4.5910	4.8104	5.2115
$q = 17$	4.1056	4.3328	4.5168	4.7781	4.9707	5.4001
$q = 18$	4.2867	4.5176	4.7264	4.9975	5.1839	5.6011
$q = 19$	4.4633	4.7065	4.9491	5.2263	5.3437	5.7193
$q = 20$	4.6387	4.9122	5.1479	5.3868	5.5574	5.9653

Table 3: Sizes, powers and size-adjusted-powers for KS and G test statistics for structural break in mean under shift in level (Equation (48)) for $n = 500$ and 1000.

n		η	KS_n^R	G_n	KS_n^W	KS_n^0	$\mathrm{KS}_n^{(b)}$		
							(b=0.1)	(b=0.5)	(b=0.9)
500	Sizes	0.0	0.147	0.065	0.068	0.048	0.025	0.021	0.015
	Powers	0.5	0.931	0.967	0.001	0.878	0.027	0.000	0.000
		1.0	0.987	0.995	0.000	0.236	0.005	0.000	0.000
		1.5	0.991	0.999	0.000	0.069	0.000	0.000	0.000
		2.0	0.991	0.998	0.000	0.040	0.000	0.000	0.000
		2.5	0.995	0.999	0.000	0.021	0.000	0.000	0.000
		3.0	0.996	0.997	0.000	0.017	0.000	0.000	0.000
	Size-adjusted -power	0.5	0.716	0.959	0.001	0.939	0.218	0.000	0.000
		1.0	0.917	0.995	0.000	0.361	0.130	0.000	0.000
		1.5	0.957	0.998	0.000	0.108	0.079	0.000	0.000
		2.0	0.975	0.997	0.000	0.050	0.040	0.000	0.000
		2.5	0.983	0.999	0.000	0.029	0.027	0.000	0.000
		3.0	0.988	0.996	0.000	0.022	0.020	0.000	0.000
1000	Sizes	0.0	0.111	0.053	0.063	0.031	0.030	0.019	0.016
	Powers	0.5	0.967	0.983	0.000	0.989	0.009	0.000	0.000
		1.0	0.987	0.996	0.000	0.839	0.001	0.000	0.000
		1.5	0.992	1.000	0.001	0.257	0.001	0.000	0.000
		2.0	0.997	0.999	0.000	0.113	0.001	0.000	0.000
		2.5	0.997	1.000	0.000	0.070	0.001	0.000	0.000
		3.0	0.999	1.000	0.000	0.039	0.002	0.000	0.000
	Size-adjusted -power	0.5	0.915	0.980	0.000	0.989	0.038	0.000	0.000
		1.0	0.978	0.994	0.000	0.828	0.009	0.000	0.000
		1.5	0.991	1.000	0.000	0.255	0.004	0.000	0.000
		2.0	0.995	0.999	0.000	0.111	0.005	0.000	0.000
		2.5	0.995	1.000	0.000	0.068	0.002	0.000	0.000
		3.0	0.996	1.000	0.000	0.039	0.002	0.000	0.000

Table 4: Sizes, powers and size-adjusted-powers for KS and G test statistics for structural break in mean under increase in α_1 (Equation (49)) for $n = 500$ and 1000.

n		α_1	KS_n^R	G_n	KS_n^W	KS_n^0	$\mathrm{KS}_n^{(b)}$		
							(b=0.1)	(b=0.5)	(b=0.9)
500	Sizes	0.0	0.084	0.043	0.020	0.044	0.020	0.003	0.002
	Powers	0.1	0.106	0.067	0.024	0.056	0.024	0.002	0.002
		0.3	0.172	0.099	0.042	0.058	0.023	0.012	0.012
		0.5	0.796	0.719	0.001	0.003	0.020	0.000	0.000
		0.7	1.000	1.000	0.000	0.000	0.000	0.000	0.000
		0.9	1.000	1.000	0.000	0.000	0.000	0.000	0.000
	Size-adjusted -powers	0.1	0.092	0.078	0.041	0.057	0.064	0.050	0.051
		0.3	0.151	0.130	0.063	0.059	0.067	0.078	0.081
		0.5	0.788	0.743	0.001	0.003	0.197	0.009	0.009
		0.7	1.000	1.000	0.000	0.000	0.000	0.000	0.000
		0.9	1.000	1.000	0.000	0.000	0.000	0.000	0.000
1000	Sizes	0.0	0.067	0.055	0.022	0.054	0.024	0.005	0.008
	Powers	0.1	0.072	0.051	0.037	0.042	0.022	0.007	0.002
		0.3	0.159	0.084	0.061	0.077	0.028	0.016	0.012
		0.5	0.826	0.745	0.001	0.003	0.024	0.002	0.001
		0.7	1.000	1.000	0.000	0.000	0.000	0.000	0.000
		0.9	1.000	1.000	0.000	0.000	0.000	0.000	0.000
	Size-adjusted -powers	0.1	0.055	0.066	0.047	0.039	0.048	0.047	0.043
		0.3	0.128	0.094	0.087	0.074	0.065	0.085	0.077
		0.5	0.816	0.758	0.002	0.003	0.169	0.007	0.010
		0.7	1.000	1.000	0.000	0.000	0.000	0.000	0.000
		0.9	1.000	1.000	0.000	0.000	0.000	0.000	0.000

Table 5: Sizes, powers and size-adjusted-powers for EKS_n^R and G_n test statistics for structural break in mean under shift in level (50) for $N = 250$ and 500.

		(i)				(ii)			
		$N = 250$		$N = 500$		$N = 250$		$N = 500$	
	$\boldsymbol{\eta}$	$\mathrm{EKS}_n^R(2)$	$\mathrm{G}_n(2)$	$\mathrm{EKS}_n^R(2)$	$\mathrm{G}_n(2)$	$\mathrm{EKS}_n^R(2)$	$\mathrm{G}_n(2)$	$\mathrm{EKS}_n^R(2)$	$\mathrm{G}_n(2)$
Size	$(0.0, 0.0)$	0.0680	0.0600	0.0330	0.0570	0.0510	0.0700	0.0400	0.0600
	$(0.5, 0.5)$	0.2220	0.0670	0.1720	0.0830	0.1970	0.0690	0.1980	0.0790
	$(1.0, 1.0)$	0.3310	0.0840	0.3300	0.0880	0.3340	0.0950	0.3150	0.0860
	$(1.5, 1.5)$	0.4690	0.1050	0.4120	0.0910	0.4620	0.1010	0.4120	0.0850
Power	$(2.0, 2.0)$	0.5250	0.0790	0.4970	0.0780	0.5030	0.0950	0.4780	0.0780
	$(2.5, 2.5)$	0.5730	0.0830	0.5360	0.0920	0.5790	0.0750	0.5340	0.0860
	$(3.0, 3.0)$	0.6320	0.0900	0.5410	0.0910	0.6130	0.0830	0.5720	0.0850
	$(3.5, 3.5)$	0.6590	0.0910	0.5970	0.0970	0.6370	0.0810	0.6010	0.0830
	$(4.0, 4.0)$	0.6830	0.0760	0.5790	0.0970	0.6670	0.1000	0.6200	0.1030
	$(4.5, 4.5)$	0.6970	0.0970	0.6350	0.0710	0.6980	0.0880	0.6080	0.0900
	$(5.0, 5.0)$	0.7210	0.1050	0.6500	0.0870	0.7260	0.0790	0.6480	0.0790
	$(0.5, 0.5)$	0.1750	0.0690	0.2300	0.1000	0.1510	0.0650	0.2860	0.0910
	$(1.5, 1.5)$	0.2540	0.0890	0.4390	0.1060	0.2650	0.0920	0.4670	0.1060
	$(1.0, 1.0)$	0.3680	0.1060	0.5370	0.1150	0.3700	0.0950	0.6500	0.1000
Size-adjusted	$(2.0, 2.0)$	0.4070	0.0800	0.6390	0.0990	0.3990	0.0880	0.7430	0.1000
-powers	$(2.5, 2.5)$	0.4390	0.0830	0.6980	0.1120	0.4360	0.0700	0.8130	0.1030
	$(3.0, 3.0)$	0.4510	0.0930	0.7190	0.1080	0.4570	0.0710	0.8830	0.1010
	$(3.5, 3.5)$	0.4740	0.0950	0.7970	0.1280	0.4810	0.0750	0.9160	0.1020
	$(4.0, 4.0)$	0.4610	0.0800	0.8160	0.1210	0.4730	0.0900	0.9490	0.1170
	$(4.5, 4.5)$	0.4620	0.0980	0.8520	0.0890	0.4740	0.0800	0.9700	0.1060
	$(5.0, 5.0)$	0.4450	0.1070	0.9110	0.1210	0.4750	0.0760	0.9850	0.0940

Table 6: Summary statistics of rates of return for ten world's major stock indices.

	Min	Max	Mean	Std Dev	Skewness	Kurtosis	ADF test		Jarque Bera test	
							Test statistic	p value	Test statistic	p value
Dow Jones	-13.8418	10.7643	0.0398	1.0571	-1.1809	28.8425	-12.3617	0.0100	83377.5652	0.0000
NYSE	-12.5972	9.5642	0.0311	1.0241	-1.3872	24.5553	-12.3846	0.0100	60798.1184	0.0000
SP Composite	-13.1761	11.2945	0.0171	0.9273	-1.8542	48.2642	-12.4516	0.0100	233268.6474	0.0000
FTSE 100	-11.5125	8.6667	0.0089	0.9820	-0.8562	14.2314	-13.7099	0.0100	20459.2869	0.0000
CAC 40	-13.0983	8.0561	0.0188	1.2074	-0.8108	10.7613	-13.7702	0.0100	11794.5197	0.0000
DAX	-13.0549	10.4143	0.0303	1.2123	-0.6684	10.5644	-13.5273	0.0100	11292.5874	0.0000
FTSE MIB	-18.5411	8.6893	0.0073	1.5253	-1.1958	14.8116	-13.6520	0.0100	22414.3775	0.0000
SSE	-8.8729	5.6036	0.0596	1.2981	-0.9444	7.2373	-13.1871	0.0100	5572.4332	0.0000
Nikkei 225	-8.2529	7.7314	0.0542	1.2664	-0.2090	5.1027	-13.2831	0.0100	2611.7850	0.0000
KOSPI 200	-7.9780	8.7548	0.0407	0.9837	-0.0893	8.2395	-12.5445	0.0100	6765.1409	0.0000

Table 7: Rejection rates for the structural break tests in the mean of conditional heteroskedasticity for the rates of return for ten major stock indices.

	KS_n^R	G_n	KS_n^W	$KS_n^{(b)}$	$KS_n^{(b)}$		
					(b=0.1)	(b=0.5)	(b=0.9)
Dow Jones	0.3595	0.2169	0.0021	0.2768	0.0000	0.0000	0.0000
NYSE	0.3462	0.2492	0.0000	0.2280	0.0000	0.0000	0.0000
SP Composite	0.3351	0.1866	0.0000	0.2455	0.0000	0.0000	0.0000
FTSE 100	0.4247	0.3208	0.0000	0.2089	0.0000	0.0000	0.0000
CAC 40	0.4883	0.3356	0.0000	0.2683	0.0000	0.0000	0.0000
DAX	0.4305	0.3908	0.0000	0.2784	0.0000	0.0000	0.0000
FTSE MIB	0.3945	0.2455	0.0000	0.0806	0.0000	0.0000	0.0000
SSE 225	0.4756	0.4883	0.0000	0.4889	0.0000	0.0000	0.0000
Nikkei 225	0.3600	0.2492	0.0000	0.2296	0.0000	0.0000	0.0000
KOSPI 220	0.3446	0.2285	0.0180	0.0048	0.0000	0.0000	0.0000

Table 8: Rejection rates for the parameter constancy tests based on the ARCH model for the rates of return for ten major stock indices.

	$KS_{r,n}^R$	$G_{r,n}$	$KS_{r,n}^W$	$KS_{r,n}^{(b)}$	$KS_{r,n}^{(b)}$		
					(b=0.1)	(b=0.5)	(b=0.9)
Dow Jones	0.2142	0.1909	0.0027	0.2269	0.1188	0.0000	0.0037
NYSE	0.2147	0.2153	0.0000	0.1893	0.1357	0.0000	0.0000
SP Composite	0.1856	0.1448	0.0005	0.1766	0.1389	0.0005	0.0000
FTSE 100	0.2582	0.2937	0.0000	0.3510	0.2566	0.0074	0.0000
CAC 40	0.3017	0.3224	0.0000	0.3420	0.1808	0.0000	0.0000
DAX	0.3070	0.3834	0.0000	0.4088	0.2497	0.0000	0.0000
FTSE MIB	0.3017	0.2450	0.0000	0.3165	0.1479	0.0000	0.0000
SSE 225	0.3436	0.4364	0.0085	0.4374	0.3054	0.0000	0.0000
Nikkei 225	0.1845	0.1379	0.0011	0.1930	0.1267	0.0000	0.0000
KOSPI 200	0.1750	0.1729	0.0175	0.1607	0.0403	0.0027	0.0154

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